



2026-27

CONCEPTS | CLINICALS | MCQS

PATHOLOGY

- From Mechanism to Diagnosis
Think Like an Examiner.
- Image-based questions &
buzzwords covered.
- General + systemic pathology
in rapid revision format.



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PATHOLOGY

CELLULAR INJURY

Cell injury → adaptation → reversible injury → irreversible injury → cell death

Most common cause of injury- Hypoxia

Most common cause of hypoxia- Ischemia

Most resistant cell to hypoxia- fibroblast

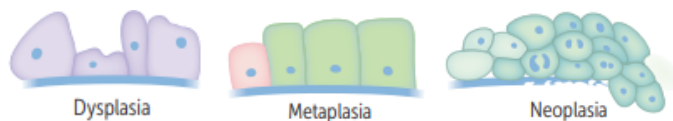
Most sensitive cell to hypoxia- neurons

Normal cells → change in cell structure → **Dysplasia**

NEETPG 2021

Normal cells → change in cell type → **Metaplasia**

Normal cells → change in cell type and cell structure → **Neoplasia**



Metaplasia- when one type of cell is replaced by another type that can handle a new kind of stress.

→ This usually happens because of constant irritation.

Example: In Barrett esophagus, stomach acid changes the lining of the lower esophagus. Squamous epithelium into columnar epithelium (contain goblet cell)

Types of cell injury

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Reversible cell injury	Irreversible cell injury
↓ ATP → due to failure for Na⁺/K⁺ pump → cellular swelling (first morphological changes) Ribosomal/polysomal detachment → ↓ protein synthesis Most common organelle affected- mitochondria Myelin figures formation (made of phospholipids and Ca⁺)	Breakdown of plasma membrane → cytosolic enzymes (eg, troponin) leak outside of cell, influx of Ca²⁺ → activation of degradative enzymes Nuclear changes- Pyknosis- nuclear chromatin condensation Karyorrhexis - nuclear fragmentation karyolysis - dissolution of nucleus AMORPHOUS DENSETIES +



NECROSIS

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1.Coagulative necrosis

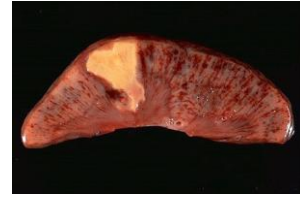
Most common type of necrosis

Due to denaturation of proteins

Seen in solid organs except the brain.

GHOST CELL seen

Preserved cellular architecture (cell outlines seen)

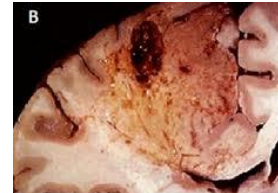


2.Liquefactive necrosis

Seen in CNS and pancreas

Lysosomal enzymes released by neutrophils that digest the tissue

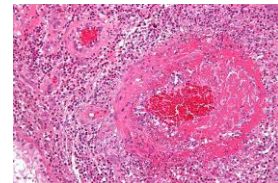
Tissue architecture not preserved



3.Fibrinoid necrosis

Associated with antigen antibody reactions

Eg→ SLE,PAN,malignant hypertension

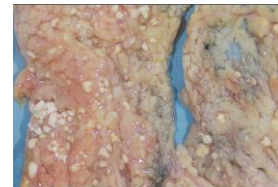


4.Fat necrosis

Seen in fatty organs- breast , peripancreatic fat

-Damaged pancreatic cells release lipase.

-Lipase breaks down fat → forms free fatty acids → bind with calcium → forming chalky white deposits



5.Caseous necrosis

Coagulative >> liquefactive

Associated with high lipid content in the cell wall

Cheeselike gross appearance

Eg- tuberculosis,histoplasma



APOPTOSIS

ATP- dependent programmed cell death

→ Can start by intrinsic pathway, extrinsic pathway or perforin/granzyme B pathway.

- All pathways activate caspases (enzymes) → cause controlled cell breakdown.

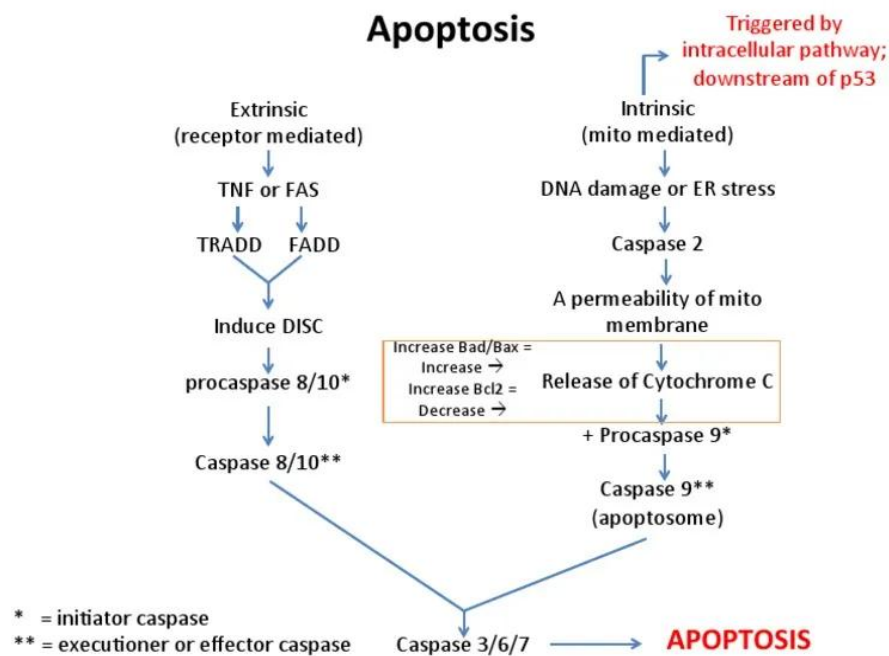
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Proapoptotic	Antiapoptotic
BAK BAX BCL-XS P53	BCL2 BCL-XL MCL 1

Mechanism of apoptosis

INITIATION- INTRINSIC AND EXTRINSIC

EXECUTION



EXECUTION

Depends on caspase 3,6 and 7

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Feature	Necroptosis	Pyroptosis	Ferroptosis
Key Mediators	RIP1, RIP3, MLKL	Caspase-1 (sometimes Caspase-4/5/11), Gasdermin D, NOD receptor	Iron, lipid peroxidation, ROS
Trigger	TNF- α pathway, failure of Caspase-8	Intracellular infection → activation of inflammasome (eg, NLRP3)	Iron overload + lipid ROS accumulation
Morphology	Cell swelling, membrane rupture, inflammation	Cell swelling, pore formation in membrane, inflammation	Small mitochondria with condensed membrane, lipid ROS damage

Calcification

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Dystrophic calcification	Metastatic calcification
Dead and degenerative tissues seen Serum Ca ⁺ normal Eg- TB Atheroma PSOMOMMA BODIES	Tissues are normal Blood calcium is ↑ Eg- Hyperparathyroidism Cancer Vitamin d intoxication

Eg- Papillary carcinoma Prolactinoma Serous cyst adenocarcinoma Serous ovarian tumor Meningioma Mesothelioma	Multiple myeloma Sarcoidosis
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Pigment and stains

Lipofuscin/wear & tear pigment

Tell tale sign of free radical injury

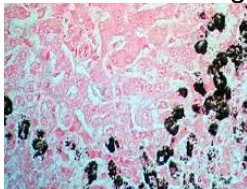
Derived by lipid peroxidation of polyunsaturated lipids of subcellular membranes

Seen in ageing, protein energy malnutrition and cancer cachexia.



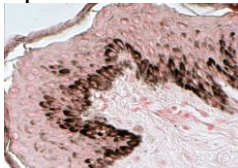
Calcium

Von kossa stain- give black color



Melanin

Special stain : Masson - Fontana

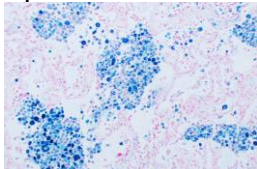


Hemosiderin

Heme derived pigment

Seen at - Local hemorrhage or bruise

Special stain: Perls Prussian blue stain



Copper

Rhodanine stain

Rubeanic stain

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Mallory hyaline/ Denk bodies

Composed of cytokeratin 8/18

Alcoholic liver disease

CELLULAR AGEING

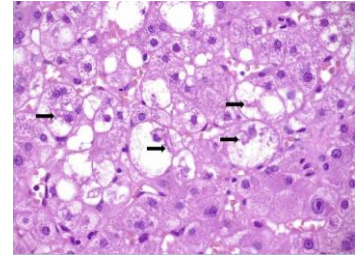
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Causes-

Free radicals injury- most common cause of ageing

Telomere shortening

Insulin resistance



Telomerase- RNA dependent DNA polymerase enzyme

Inhibits cellular aging (Immortality gene)

Produces more telomeres

High telomerase activity : Cancer cells, stem cells, germ cells

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Sirtuins-

NAD dependent protein deacetylases → ↑longevity

Role in- Diabetes mellitus, cancer, cellular aging

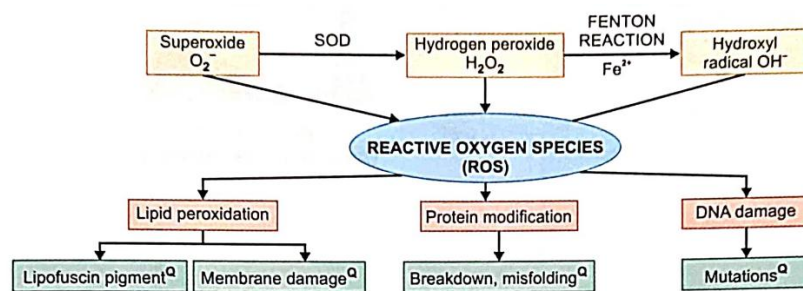
Calorie restriction - ↑lifespan (D/t ↓cellular aging)

PREMATURE AGEING

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Werner syndrome aka adult progeria	Hutchinson gilford syndrome
Occur in adulthood DNA helicase defect	Occur in childhood LAMIN A gene defect

FREE RADICAL INJURY



*SOD-Superoxide dismutase

Most potent free radical- OH-

Clinical correlation

Superoxide dismutase(SOD) is of two types-

-Cytoplasmic

-Mitochondrial

SOD 1 gene mutation can cause ALS (Amyotrophic Lateral sclerosis)

ALS is both upper and lower motor neuron disorder

BUNINA BODIES- Microscope studies show

Fenton reaction

The Fenton reaction is a chemical reaction in which ferrous iron (Fe^{2+}) reacts with hydrogen peroxide (H_2O_2) to produce highly reactive hydroxyl radicals ($\cdot\text{OH}$).

Q) Which of the following pigments are involved in free radical injury?

- A) Lipofuscin
- B) Melanin
- C) Bilirubin
- D) Hematin

ACUTE PHASE REACTANT

POSITIVE APR	NEGATIVE APR
Mnemonic: "C-FERRITIN CASH" C = C-reactive protein (CRP) FERRITIN = Ferritin C = Complement A = Alpha-1 antitrypsin S = Serum amyloid A H = Hepcidin	Albumin Transferrin Transthyretin Retinol binding protein

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$\text{TNF-}\alpha \rightarrow$ no appetite
 $\text{TGF-}\alpha \rightarrow$ gastric disease (menetrier disease)
 $\text{TGF-}\beta \rightarrow$ most fibrogenic cytokine

INTERLEUKINES

$\text{IL-1, TNF-}\alpha$ - FEVER
 $\text{IL-10, TGF-}\beta$ - ANTI INFLAMMATORY
 IL-5 - EOSINOPHIL ACTIVATION
 IL-8 - NEUTROPHIL RECRUITMENT

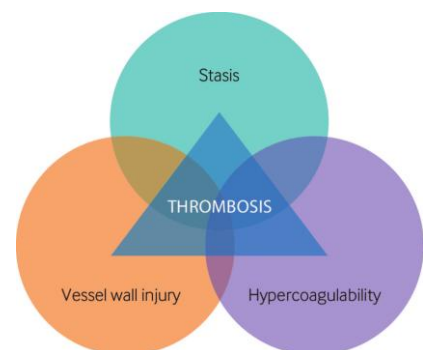
ACUTE INFLAMMATION

VIRCHOW TRIAD

Vascular events

Vasoconstriction $\rightarrow$ vasodilation $\rightarrow$ increased vascular permeability $\rightarrow$ stasis

NEETPG 2024



Cellular events

FMGE 2021

Margination	Cells move towards margin	
Rolling	Attachment and detachment to endothelium	Mediated by selectin 3 types of selectin P(platelet and endothelial cell)- bind to sialyl lewis X E(endothelial cell)- bind to sialyl lewis X L(leucocytes)-bind to GLYCAM-1/CD 34
Adhesion	WBCs stick firmly to the endothelium using integrins.	Mediated by integrin β1 bind to VCAM 1 β2 bind to ICAM 1
Transmigration	WBCs squeeze between endothelial cells to exit the blood vessel	Mediated by PECAM/CD31
Chemotaxis	Unidirectional movement of WBC towards bacteria	L- LTB4 I- IL8 C- C5a
Opsonisation	Pathogens are coated with opsonins	IgG(most important opsonin) Complement opsonin- C3b,C4b,C5b
Phagocytosis	WBCs engulf and destroy the pathogen.	Oxygen dependent killing (most common)

Neutrophil extracellular traps (NETs)

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A web-like (fibrillar) mesh made by **neutrophils** at the infection site.

-Contains high concentration of antimicrobial substances.

Need: Arginine is required for NET formation.

LEUCOCYTE ADHESION DEFECT

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LAD I	Defect in β2 Integrin	Delayed separation of umbilical cord
LAD II Trick→ II- second- S	Defect in Sialyl lewis X	Bombay blood group of patient

INTERFERONS

Interferons	Source	function
INF-α	L- leucocyte	Antiviral property
INF-β	F-fibroblast	Treatment for multiple sclerosis

INF-gamma

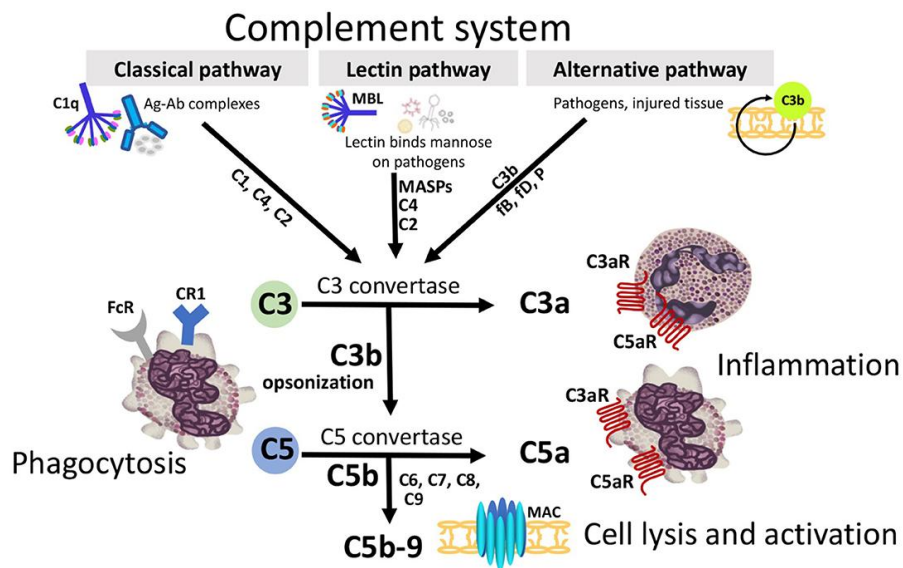
T- Th1

Granuloma formation

C3b—opsonization.

C3a, C4a, C5a—anaphylaxis.

C5a—neutrophil chemotaxis.



Deficiency	Disease
Most common complement deficiency	C2
C1,2,4 deficiency	SLE
MAC deficiency (C5b – C9)	Capsulated organism <i>Neisseria</i> infection
C1 inhibitor deficiency ↓ Controls activation of C1	HAE Hereditary angioEdema

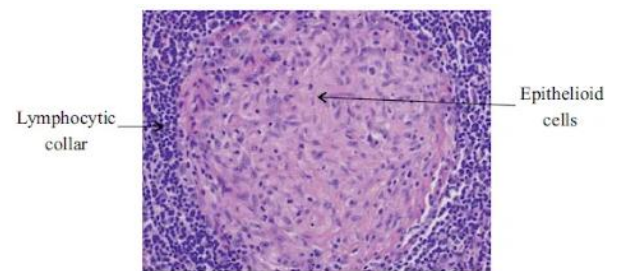
Deficiency	Disease
CD55 (break C3 conv)	Paroxysmal nocturnal hemoglobinuria
CD59 (break C5 conv)	
Factor H	Atypical HUS (Hemolytic uremic syndrome)
Factor I	
CD46	
Factor H defect in ophthalmology	ARMD (Age related macular degeneration)

Macrophages in different organs

Brain - microglia cells

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Liver - kupffer cells
Lungs - alveolar cells
Spleen - littoral cells
Placenta - Hofbauer cells



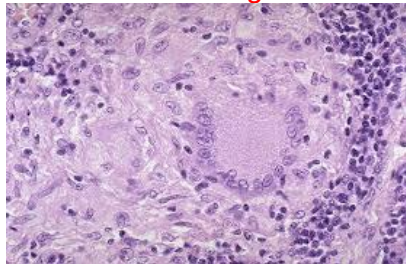
Granuloma

Th1 releases interferon gamma → granuloma

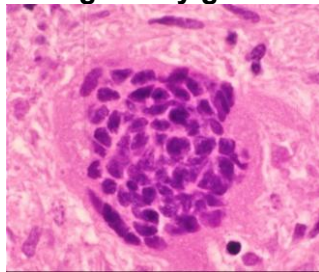
Types of giant cell

Langhans giant cell

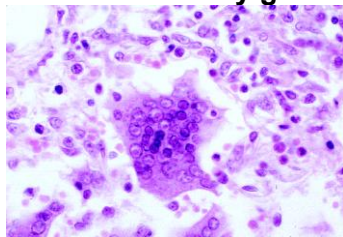
- Seen in TB
- horse shoe arrangement of nuclei



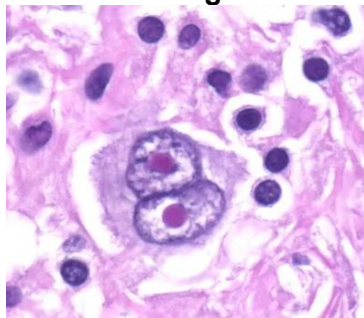
Foreign body giant cell- Haphazard arrangement of nuclei ,Seen in Xanthoma



Warthin finkeldey giant cell- seen in measles



Reed sternberg cells- Hodgkin's lymphoma



TB → caseating granuloma > non caseating granuloma

Sarcoidosis → non caseating > caseating granuloma

Cat scratch disease, LGV → Stellate granuloma

Q fever, allopurinol → Doughnut granuloma

Malaria falciparum → Durck granuloma

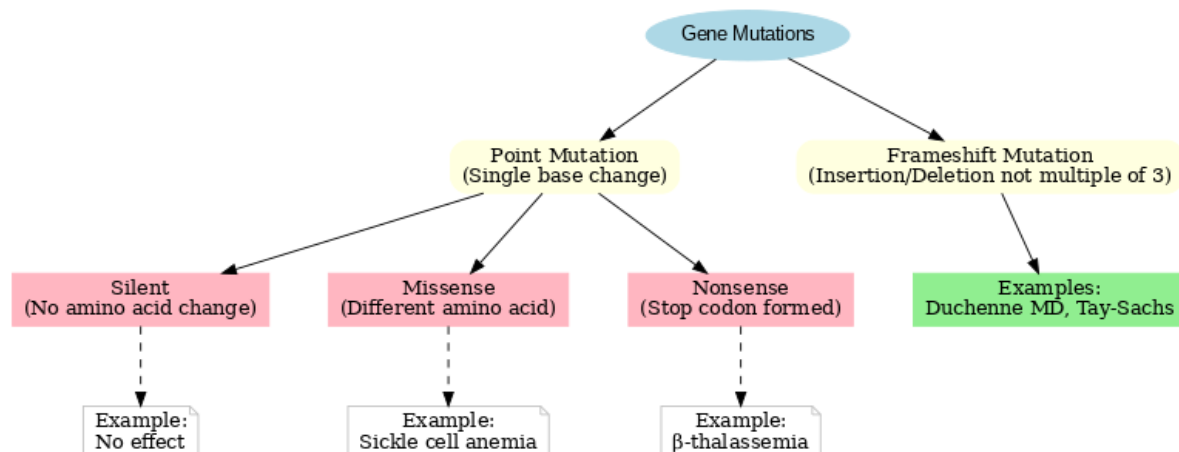
Events of wound healing

Events of wound healing

- Learn the topic in terms of days – 0, 1, 2, 5, 21 days (3 weeks)

Day	Event
Day 0	Platelets and Fibrin forms Blood clot
Day 1	• Blood clot remains • Neutrophils (acute inflammation)
Day 2	• Blood clot remains • Neutrophils (acute inflammation) • Thin epithelial layer forms
Day 3	• Chronic inflammation starts • Macrophages will come • Granulation tissue begins (blood vessel formation starts) • Beginning of collagen (Type III) formation ◦ At the margins
PYC (possibly Day 5)	• Maximum granulation tissue (Maximum neovascularization) • Collagen bridging starts
Day 21 (Week 3)	• Maximum collagen (Type I)
Day 21	Maximum collagen (Type I)

GENE MUTATION



Tumor showing bony metastasis

Prostate tumor - osteoblastic

Breast and lungs tumor - osteolytic > osteoblastic
Kidney tumor - osteolytic

MENDELIAN DISORDER

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1.AUTOSOMAL DOMINANT

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Trick to remember →

Seen in every generation - Dominant- Doesn't skip generation

Both sexes affected equally

Father to son transmission → autosomal

Eg- Father's HAVE Very DOminant Genes

F → FAP

H → HOCM, Huntington

HAVE → Hereditary spherocytosis, HNPCC,ADPKD,VHL, vWD

Dominant → Dystonia (muscle),Osteogenesis imperfect,Marfan syndrome,intermittent porphyria,Noonan syndrome, NF1, NF2,Achondroplasia,Tuberous sclerosis

Tuberous sclerosis

TSC gene defect

TSC 1 on chr9 and TSC 2 on chr16

Protein made by TSC 1 is Hamartin and TSC 2 tuberlin,there is multiple hamartoma.

Hypopigmentation regions known as Ash Leaf macules.

2.AUTOSOMAL RECESSIVE

Look for skipped generations

Disease seen in siblings, but parents unaffected.

Both sexes are equally affected.

If consanguineous marriage is mentioned → think AR.

1 in 4 rule if both parents are carriers.

Eg-

"SPAT COW HX"

S → Sickle cell anemia

P → Phenylketonuria

A → Alkaptonuria

T → Thalassemia

C → Cystic fibrosis

O → α -1 antitrypsin deficiency

W → Wilson disease

H → Hemochromatosis / Homocystinuria / Hemophilia C

X → Xeroderma pigmentosum

3.X-LINKED RECESSIVE

No male-to-male transmission.

Affected father → all daughters affected, no sons.

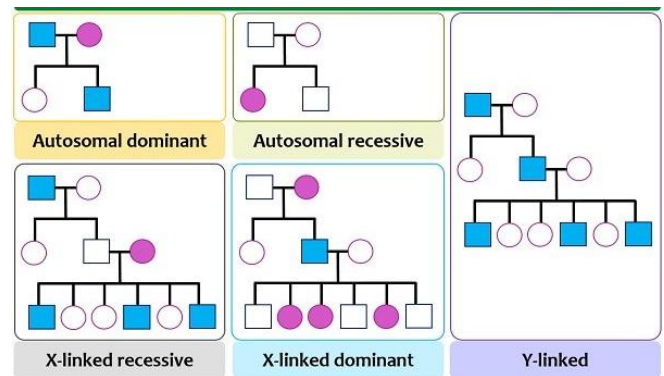
Eg-

Hunter disease
 G6PD
 Lesch nyhan disease
 Fabry's disease
 Wiskott aldrich syndrome
 Hemophilia A&B
 Duchene muscular dystrophy

4.X-LINKED DOMINANT

No male-to-male transmission.
 Affected mother (carrier) → sons more affected.

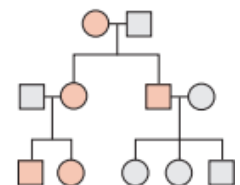
Eg- RAVI
 R-rett syndrome
 A-alport disease
 V-vitamin d resistant rickets
 I- incontinentia pigmenti



Mitochondrial inheritance

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Transmitted only through the mother. All offspring of affected females may show signs of disease.



Memory Tricks:

X linked Dominant = Daughters → Father affects all daughters.
 X linked Recessive = Rare sons → More boys affected, skips genes.

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PRADER WILLI	ANGELMAN SYNDROME
Snorp gene involve Paternal deletion (maternal gene imprinted/silent)	UBE3A gene involve Maternal deletion (paternal gene imprinted/silent)
Hyperphagia → obesity - Short stature - Hypogonadism - Intellectual disability	- Frequent laughter, happy demeanor - Ataxia, seizures - Severe intellectual disability - Inappropriate smiling/hand-flapping

SOUTHERN BLOTTING - DNA
 NORTHERN BLOTTING - RNA
 WESTERN BLOTTING - Protein

PCR

Steps → Denaturation,Annealing,Extension

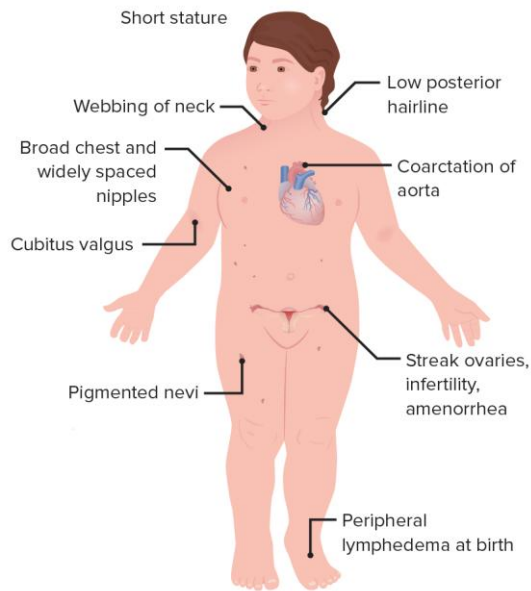
Turner syndrome

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45XO

Most common cause of primary amenorrhea

Most common bicuspid aortic valve



Trinucleotide repeats

Fragile x syndrome	Myotonic dystrophy	Friedreich ataxia
CGG repeats FMR 1 gene involve Chromosome Xq Sharman paradox + → with each generation no of repeats and severity ↑ C/F- Long face, ears, mandible Macro-orchidism Mental retardation	CTG repeats C- christmas tree cataract T- testicular atrophy G-hypogammaglobulinemia HATCHET FACIES	GAA repeats

Karyotyping-Clinical application : Chromosomal disorders.

Banding/Stain : Giemsa/G-banding (m/c).

Chemical use is colchicine

Arrested in metaphase stage

Fixative - carnoy fixative

DNA repair defects

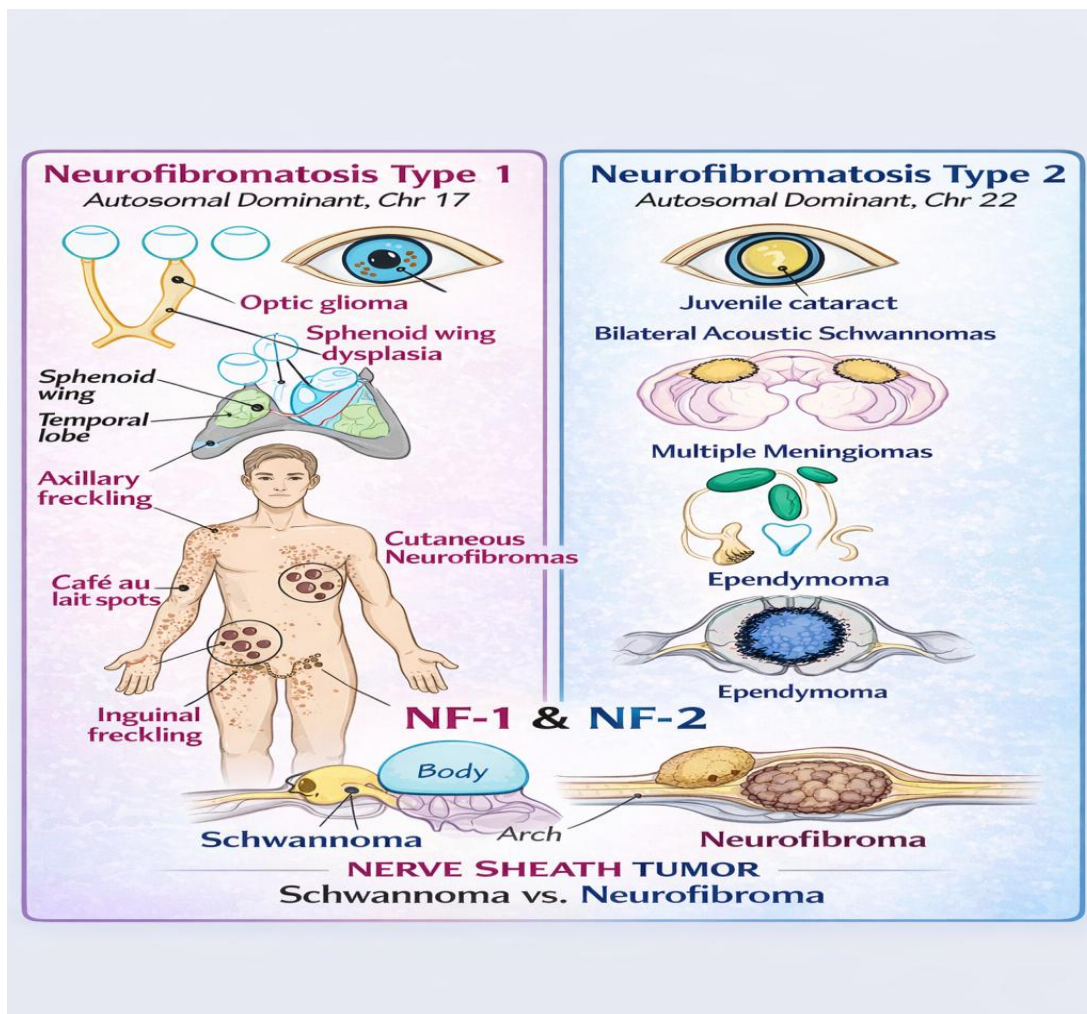
NEETPG 2024

NER defects - nucleotide excision repair defect → xeroderma pigmentosa

MMR defects - mismatch repair → Lynch syndrome (colon cancer ,endometrial cancer ,ovarian cancer)

Homologous recombination defect→ bloom syndrome,Ataxia telangiectasia,fanconi anemia

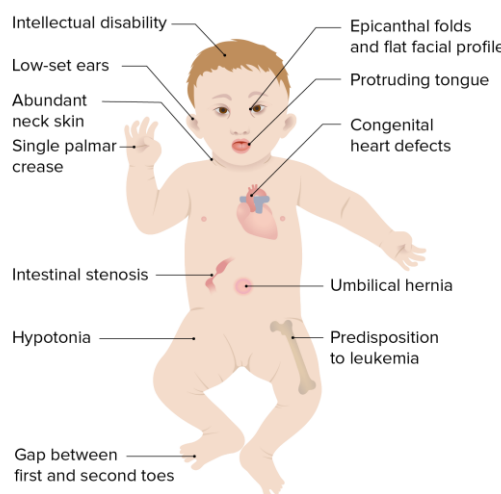
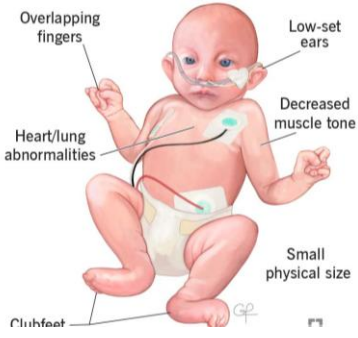
Neurofibromatosis



Autosomal trisomies

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Down syndrome (trisomy 21)	Edwards syndrome (trisomy 18)	Patau syndrome (trisomy 13)
C/F: 94 % is due to meiotic non disjunction of chromosome 21 in ova.	Findings: PRINCE Edward→ -Prominent occiput -Rocker-bottom feet - Intellectual disability - Nondisjunction -Clenched fists with overlapping fingers -low-set Ears congenital heart disease (eg, VSD)	Findings: -severe intellectual disability -Rocker bottom feet - - Microphthalmia - - Microcephaly -Polydactyly -Cleft lip/palate

 Alzheimer disease (chromosome 21 codes for amyloid precursor protein). ALL is most common <3yr- AML-M7 is common Endocardial cushion defect is most common cardiac disease		 Aplasia cutis 
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ONCOGENES

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Kras	Colon ca.
Hras	Bladder ca.
Nras	melenoma
C-kit	Gist
RET	Medullary ca. of thyroid
ABL	CML
JAK	Polycythemia vera
C-myc L-myc N-myc	Burkitt lymphoma Small cell lung cancer neuroblastoma

TUMOR SUPPRESSOR GENE

FMGE 2025,23,NEETPG 2024

Gene	Protein	Tumors
NF-1	Neurofibromin	Optic nerve glioma Neurofibromas
NF-2	Merlin	Meningioma

		Schwannoma
Rb	Retinoblastoma protein	Retinoblastoma Osteosarcoma
BRCA-1	BRCA-1	Ovarian cancer Breast cancer
BRCA-2	BRCA-2	Prostate cancer
Tp53	P53 protein	Lifraumeni syndrome
PTEN	Phosphate and tensin homologue	Endometrial cancer Prostate cancer
VHL	VHL protein	Clear cell RCC
CDH-1	E-cadherin	Invasive lobular carcinoma breast Diffuse gastric adenocarcinoma
STK-11	Liver kinase B1	Peutz-jeghers syndrome

Cyclins and CDK

Cyclins	CDK
D	4,6
E	2
A	2
B	1

Warburg effect

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Normal cells: Use aerobic respiration → Glucose → Glycolysis → Pyruvate → Mitochondria
→ Oxidative phosphorylation (36 ATP per glucose)

Cancer cells: Even when oxygen is present, they prefer aerobic glycolysis → Glucose →
Glycolysis → Lactate (only 2 ATP per glucose)

This is called the Warburg Effect (aerobic glycolysis)

HEMOCHROMATOSIS

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Autosomal recessive

Mutation in HFE gene Low hepcidin—> increase in iron absorption

Stain used - Prussian blue

Treatment: Repeated phlebotomy Iron (Fe) chelation with Deferoxamine, Deferiprone

CHEMICAL AND MICROBIAL CAUSING CANCER

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Benzene → AML

Benzenidine → urinary bladder cancer

Asbestos → most common → adenocarcinoma of lung

→ specific cancer → mesothelioma → lung

Vinyl chloride, Thorotrast, Arsenic → Angiosarcoma

HBV → HCC

Helicobacter → most common → gastric adenocarcinoma

HHV8 → castleman disease, Kaposi sarcoma, primary effusion lymphoma

Aspergillus flavus → Aflatoxin → HCC

Clonorchis/Optisthorchis → Cholangiocarcinoma

EBV → most common → gastric carcinoma

Schistosoma haematobium → SCC of urinary bladder

HTLV → Adult T cell leukemia lymphoma

HPV → low risk HPV → 6, 11 → warts → condyloma acuminata

→ high risk HPV → 16, 18 → SCC → cervix penile, anal, esophagus, larynx

Tumor markers

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β HCG → choriocarcinoma

Calcitonin → medullary carcinoma of thyroid

5HIAA → carcinoid tumor

Catecholamines → Pheochromocytoma

Alpha - fetoprotein → Yolk sac tumor, HCC, Hepatoblastoma

CA 125 → Surface epithelial ovarian cancer

CA 19-9 → Pancreatic cancer

PSA → Prostate cancer

NMP22 → recurrent bladder cancer

Calretinin → mesothelioma

β2 microglobulin → multiple myeloma

PARANEOPLASTIC SYNDROME

Squamous cell cancer	Hypercalcemia
Small cell cancer	Lambert eaton syndrome Acanthosis nigricans SIADH Cushing syndrome
Fibrosarcoma	Hypoglycemia
Thymoma	Pure red cell aplasia Myasthenia gravis

IHC markers

Marker	Tumor
Cytokeratin	Carcinoma
HMB 45	MELANOMA
S100	MELANOMA, LCH
VIMENTIN	SARCOMA

TRANSPLANT REJECTION

FMGE 2021

Autograft	Tissue transplanted within the same person
Allograft	Tissue transplanted between two genetically different individuals of the same species
Isograft	Tissue transplanted between genetically identical individuals
Xenograft	Tissue transplanted between different species

ISOTYPE SWITCHING

B cells have IgM and IgD antibodies

Conversion of IgM and IgD into IgG,A,M,D,E called isotype switching

It needs - IL-4 and CD-40

Humoral immunity

IgG (max. Concentration)	General, most abundant, crosses placenta
IgA	At mucosa, in secretions
IgM	Massive, Made first
IgD	-
IgE ((min. Concentration)	Eosinophils, allergy

TYPES OF HYPERSENSITIVITY

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Type 1 (immediate)	IgE-mediated → Mast cells & basophils release histamine, leukotrienes	Eg- Allergies Bronchial asthma Drugs- penicillin	A- allergy
Type 2 (cytotoxic)	IgG or IgM binds to cell surface antigens → Complement activation → Cell lysis	Eg- MB GRIP Myasthenia gravis Blood transfusion reaction Graves disease and goodpasture RHD/Rh incompatibility ITP Pemphigus and Pernicious anemia	C- cells destroyed
Type 3 (immune complex)	Antigen-antibody (IgG/IgM) complexes deposit in tissues → Complement → Inflammation	Eg- SLE HSP PSGN, Polyarthritis nodosa	I- immune complex deposition
Type 4 (delayed)	T-cell mediated (CD4+ Th1 → macrophages OR CD8+ cytotoxic T cells)	Eg - MCG Multiple sclerosis Mantoux test Granuloma Contact dermatitis	D- delayed

IMMUNO DEFICIENCY DISORDERS

1. DI- GEORGE SYNDROME-

FMGE 2025,21

22q11 microdeletion (T cell disorder)

Developmental failure of 3rd and 4th pharyngeal pouch

Lack of thymus, hypocalcemia (lack of parathyroid), defect in heart and face.

2. SEVERE COMBINED IMMUNODEFICIENCY DISORDER (SCID)

ADA ↓, more prone to infections (bacterial, viral, fungal)

Note: ADA ↑ in TB

3. BRUTONS HYPOGAMMAGLOBULINEMIA

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Defect in BTK, a tyrosine kinase gene → no B-cell maturation

Defective opsonisation- leads to recurrent bacterial infections

Seen in boys

4. WISKOTT ALDRICH SYNDROME (WAS)

NEETPG 2024

Mutation of WASP gene - Say BIE to WAS

Traid

B- Bleeding due to thrombocytopenia

I- Infections due to leukopenia

E- Eczema

5. CHEDIAK HIGASHI SYNDROME

LYST gene defect

C- CNS abnormalities

HE- hemorrhage

DI- decreased immunity

A- albinism

K-coarse cytoplasmic granules

6. CHRONIC GRANULOMATOUS DISEASE

FMGE 2024,2026

PHOX gene defect, NADPH oxidase deficiency

Catalase positive infections seen in patients

Screening test- NBT (nitro blue tetrazolium)

Confirmatory- DHR (dihydrorhodamine test)

Frequently asked question:

Q. A 19-month-old child presents with recurrent infections. Advanced studies show a defect in neutrophil adhesion. Further, the mother gives a history of delayed umbilical cord stump shedding after birth. The child's defect is most commonly attached to?

a. LAD 1

b. LAD 2

c. BRUTON'S Hypogammaglobulinemia

d. SCID

Q.A 10-year-old child presents with repeated bleeding episodes, delayed milestones as well as hypopigmentation. What is your likely diagnosis?

- a. Defect in selectins
- b. Defect in phagocytosis
- c. Defect in opsonization
- d. Defect in integrins

AMYLOIDOSIS

FMGE 2025,23,21,NEETPG 2024

Misfolded proteins get deposited outside cells in an abnormal **β -pleated sheet** form

Primary amyloidosis → AL → Ig of light chain → seen in multiple myeloma

Secondary amyloidosis

Hemodialysis → A β 2

Alzheimer disease → A β

Medullary carcinoma of thyroid → Acal

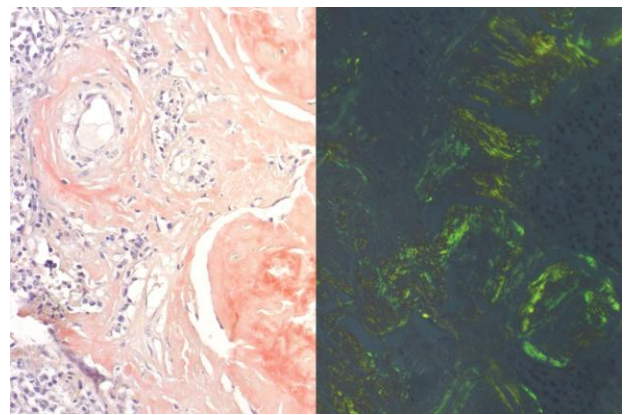
Gross stain - Lugol's iodine

Congo red stain → light microscopy → salmon pink

→ polarising microscope → apple green birefringence

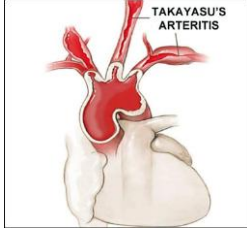
Xray crystallography- β pleated sheets

Sample collection - abdominal fat aspiration



VASCULITIS

Large cell vasculitis

TEMPORAL ARTERITIS / GIANT CELL ARTERITIS 	TAKAYASU ARTERITIS / PULSELESS DISEASE 
Age group >50 yr	Age group < 50 yrs
Most common artery affected → superficial temporal artery Common symptom → U/L headache Most specific symptoms → jaw claudication Most dangerous symptoms → loss of vision	Most common artery affected subclavian artery Upper limb pulse - weak Lower limb pulse - normal
M/E - giant cell and granuloma , fragmentation of internal elastic lamina Dx- Biopsy	
Trt- steroids	

Medium vessel vasculitis

POLYARTHRITIS NODOSA

Hbsag + seen in 30% of cases

Most common vessel involve → renal vessel → Flea-bitten kidney & not associated with glomerulonephritis

Vessel spared → pulmonary vessel

Microscopic examination → Fibrinoid necrosis → segmental and transmural

KAWASAKI DISEASE

Age <5yr

+4/5 criteria should be present

C- Conjunctivitis

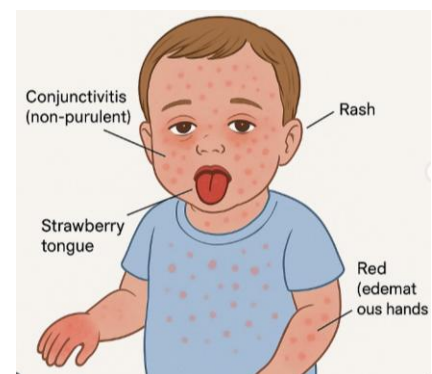
R- Rashes

E- Edema

A- Lymphadenopathy (unilateral and cervical)

M- Mucosal involvement (strawberry tongue)

Complications- Myocardial Infarction



BUERGER'S DISEASE/THROMBOANGITIS OBLITERANS

Associated with smoking
 Involve → artery, veins, nerves
 C/F → Intermittent claudication → Leg pain when walking/exercising, relieved by rest
 Rest pain → Severe pain even without activity (late stage)
 Inflammation & clot (thrombus) → Narrows or blocks the artery
 Result → Reduced / blocked blood supply to tissues



Small vessel vasculitis

HSP (Henoch Schonlein Purpura)

Platelet count → normal or increased
 Seen in children
 IgA mediated

FMGE 2024, NEETPG 2021

C/F- "PALP"

Purpuric rash

Arthralgia

Lower Abdominal pain

Proteinuria/Hematuria (most common)

GRANULOMATOSIS WITH POLYANGITIS/WEGENER GRANULOMATOSIS

Associated with c-ANCA (PR3-ANCA, anti-proteinase 3)

Causes necrotizing granulomatous inflammation and vasculitis in affected organs

C/F-

Upper respiratory → Recurrent sinusitis, Recurrent otitis, saddle nose deformity, strawberry gums

Lower airway obstruction → Hemoptysis, subglottic stenosis

Renal - AKI, glomerulonephritis

MICROSCOPIC POLYANGITIS

NEETPG 2024

P-ANCA +

No granulomas, Leucocytoclastic vasculitis+

Necrotizing vasculitis commonly involves lung, kidneys and skin with pauci-immune glomerulonephritis and palpable purpura.

EOSINOPHILIC GRANULOMATOSIS WITH POLYANGITIS/CHURG STRAUSS SYNDROME

↑ IgE level

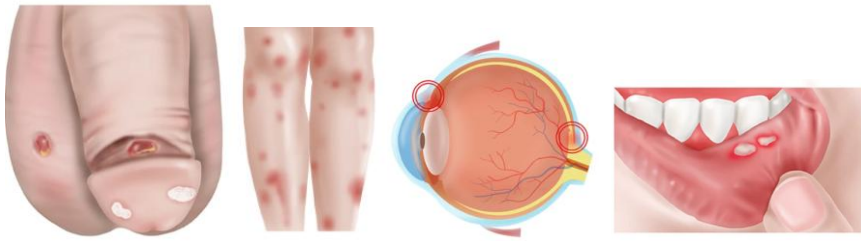
Asthma, sinusitis, skin nodules or purpura, peripheral neuropathy +

BECHET SYNDROME

Associated with HLA-B51

Recurrent aphthous ulcers, genital ulcerations, uveitis, erythema nodosum

Genital ulcer Erythema nodosum Uveitis Mouth ulcer



Capillary hemangioma 	Cavernous hemangioma 
Associated with strawberry hemangioma It occurs on skin and superficial sites Disappear by 7 yrs of age	Dilated vessel seen Most common benign tumor in the liver

Cardiac enzymes

Cardiac enzymes	Start time	Full time
Myoglobin (earliest)	2 hr	2 hrs
Troponin T and I	2-4 hr	7-10 days
CK-MB	2-4 hr	48-72 hrs
LDH-1 (last)	24 hr	After 10 days

Earliest- heart fatty acid binding protein
 LDH 2 >> LDH 1 → In Normal patient
 LDH 1 >> LDH 2 → in MI patient

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FMGE 2025,23

STAIN- TPTC → AFFECTED HEART LOOKS PALE

AUTOPSY FINDINGS IN MYOCARDIAL INFARCTIONS

FMGE 2021

Timings of death	Gross	Light microscopy
Reversible cell injury 20–30 minutes	-	-
Irreversible cell injury	-	→ vacuolation of cell → waviness of cardiac fibers

30 min - 4 hrs		
4-12 hrs	Occasionally dark mottling	Coagulative necrosis
12 - 24 hrs	Dark mottling	-Coagulative necrosis -Contraction band necrosis -Neutrophils
1-3 day		Coagulative necrosis, interstitial infiltrate of neutrophils
3-7 days		Neutrophils replaced by macrophages
7-10 days		Early granulation tissue
>2 months		Scar formation

RHEUMATIC HEART DISEASE

Acute - mitral regurgitation → pressure builds in left atrium →

Mccallums patch

Chronic - mitral stenosis → fish mouth and button hole deformity

Aschoff bodies

Most characteristic lesion in RHD

Anitschkow cell/caterpillar cells : Slender wavy ribbon like nuclei (In Aschoff bodies)

NEETPG 2021



Revised jones criteria

1.MAJOR CRITERIA- PASS Everyone

P- Pancarditis

A- Arthritis

S- Subcutaneous nodules

S- Sydenham chorea

E- Erythema marginatum

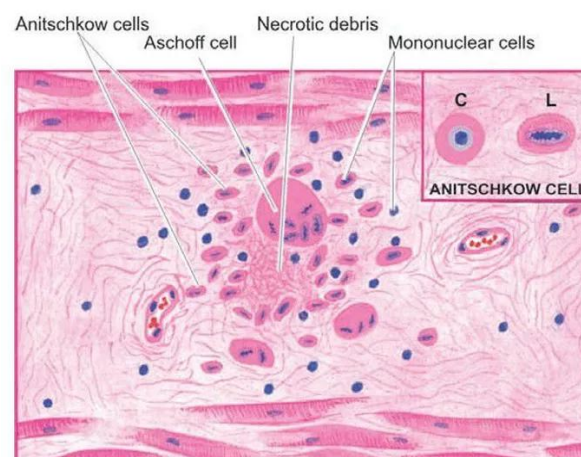
2.MINOR CRITERIA -FAIL everyone

F- Fever

A- Arthralgia

I- increased PR interval

L- Lab test ESR and CRP increased



Diagnosis - 2 major or 1 major + 2 minor

ENDOCARDITIS

Infective endocarditis	Rheumatic heart disease	Non bacterial thrombotic endocarditis	Libman sach's endocarditis
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Infection (bacteria – Staph aureus, Strep viridans, Enterococci)	Autoimmune reaction after group A β -hemolytic strep pharyngitis	Hypercoagulable state (cancer, sepsis, DIC)	SLE (immune complex deposition)
Large, friable, destructive vegetations	Small, warty vegetations along line of closure	Small, sterile vegetations	Small to medium vegetations on both sides of valve leaflets
Aortic & Mitral (Left-sided, except IV drug users → Tricuspid)	Mitral valve most common ("fish-mouth" stenosis)	Usually Mitral valve	Mitral & Tricuspid valves
			

INFECTIVE ENDOCARDITIS

Infection of the inner lining of the heart (endocardium), usually **heart valves**.

Community acquired infection

Infants - staphylococcus aureus

Adults - streptococcus

Nosocomial : staph aureus

Special association

Prosthetic valves → Staph epidermidis

Culture negative → Coxiella, Bartonella

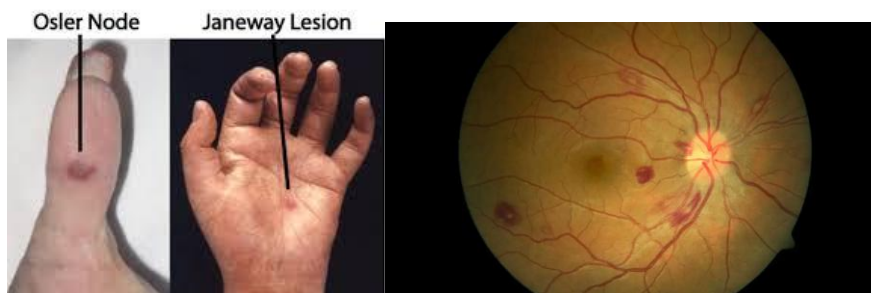
IV drug use → Staph aureus, Pseudomonas, Candida

Modified duke criteria

Criteria	Details
Major Criteria	I. Positive blood cultures for IE 1. Two separate blood cultures positive for typical organisms: Viridans streptococci, Staphylococcus aureus, HACEK group, or community-acquired enterococci (without primary focus) 2. Persistently positive blood cultures drawn >12 hrs apart OR all of 3, or

	majority of 4, blood cultures drawn ≥ 1 hr apart 3. Single positive blood culture for Coxiella burnetii II. Evidence of endocardial involvement 1. Positive echocardiogram for IE 2. Oscillating intracardiac mass on valve/supporting structure 3. Abscess 4. New partial dehiscence of prosthetic valve 5. New valvular regurgitation (not just change in murmur)
Minor criteria	1. Predisposing heart condition or intravenous drug use 2. Fever $\geq 38^{\circ}\text{C}$ (100.4°F) 3. Vascular phenomena: arterial emboli, Janeway lesions 4. Immunologic phenomena: glomerulonephritis, Osler's nodes, Roth spots, rheumatoid factor 5. Microbiologic evidence: positive blood culture not meeting major criteria, or serology consistent with IE

ROTH SPOTS



Diagnosis is made when:

2 Major criteria, OR 1 Major + 3 Minor, OR 5 Minor criteria.

LUNG CANCER

NEETPG 2023,24

features	Squamous cell cancer	Adenocarcinoma	Small cell cancer	Large cell cancer
Smoking	+	-	+	-
Site	Central	Peripheral	Central	Peripheral
Genetics	loss of Chr3 P53 gene	Gain of function EGFR1 ALK	Loss of Chr3	
Associated syndromes	Hypercalcemia	Migratory thrombophlebitis	Lambert eaton syndrome SIADH Cushing syndrome Most aggressive lung cancer	Gynaecomastia
IHC markers	P40,P63	NAPSIN A,MUC 1	Synaptophysin	

			Bombesin	
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TYPES OF EMPHYSEMA

Centriacinar	Panacinar	Paraseptal	Irregular emphysema
Most common in smokers	Most common in $\alpha 1$ antitrypsin deficiency	Can cause spontaneous pneumothorax	Most common histologically
Alveolar ducts and sacs are spared	Respiratory bronchioles, alveolar ducts and sacs are spared	Respiratory bronchioles are spared	Irregular movement

SARCOIDOSIS

FMGE 2023

Noncaseating granulomas in multiple organs

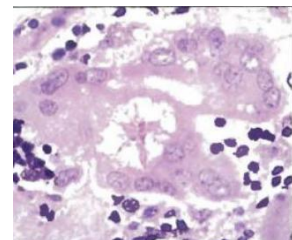
↑ serum ACE levels, and ↑ CD4/CD8 ratio in bronchoalveolar lavage fluid.

Characteristic stellate inclusions ('asteroid bodies') seen

CXR shows bilateral adenopathy and coarse reticular opacities

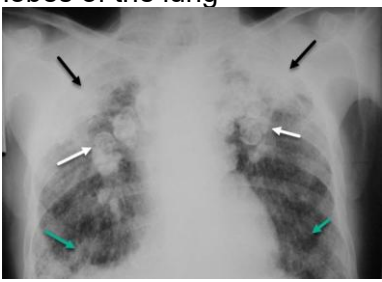
Clinical features

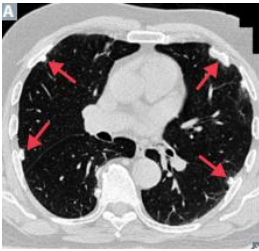
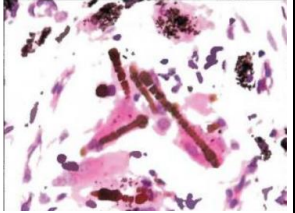
1. Dyspnea or cough (most common presenting symptom)
2. Hypercalcemia, ↑ vit D
3. Ocular uveitis
5. Skin changes (eg, lupus pernio, erythema nodosum)



PNEUMOCONIOSES

FMGE 2025,23,21,NEETPG 2023,24

ENTITY	EXPOSURE	PATHOLOGIC FINDINGS	COMMENTS
SILICOSIS	Silica; seen in sandblasters and silica miners	Fibrotic nodules in upper lobes of the lung 	↑ risk for TB and cancer

		"Eggshell" calcification of hilar lymph nodes on CXR.	
ASBESTOSIS	construction workers, plumbers, and shipyard workers	-Fibrosis of lung and pleura (plaques) -Risk for lung carcinoma and mesothelioma Affects lower lobes "Ivory white," calcified, supradiaphragmatic and pleural plaques are pathognomonic. 	Asbestos bodies  Most common - lung adenocarcinoma Most specific- lung mesothelioma
BERYLLOSI	miners and workers in the aerospace industry	Granulomatous (non caseating)	
COAL WORKER PNEUMOCONIOSES	Carbon dust; seen in coal miners	Macrophages laden with carbon, inflammation, fibrosis.	↑ risk of Caplan syndrome.

Barrett's Esophagus

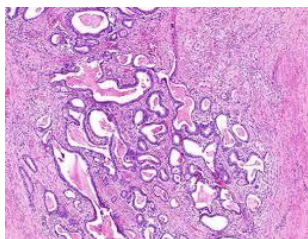
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Squamous epithelium changes into intestinal columnar epithelium

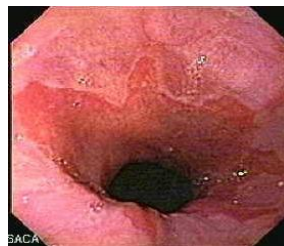
Type of metaplasia

Hallmark cell - Goblet cell

Stain for goblet cell- alcian blue stain



adenocarcinoma



ULCERS

GASTRIC ULCER	DUODENAL ULCER
Site- lesser curvature	More common

Most common complication - bleeding → left gastric artery
Pain after meal

Site- first part of duodenum
Pain relieved by taking food
Most common complication- bleeding → gastro duodenal artery

MENETRIER DISEASE

Common in males

Increase in TGF α

Clinical - **Protein losing enteropathy**

- protein ↓ causing edema

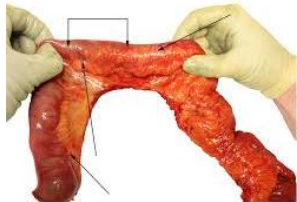
Gross- giant cerebriform rugae

Microscopy - Foveolar cell hyperplasia



CROHN DISEASE vs ULCERATIVE COLITIS

FMGE 26

Crohn disease	Ulcerative colitis
Any part of GI tract (mouth → anus), commonly terminal ileum	Limited to colon & rectum
Skip lesions (patchy, segmental)	Continuous from rectum upwards
Transmural (full-thickness), cobblestone mucosa, creeping fat 	Mucosal & submucosal only
Non-caseating granulomas (seen in ~50%)	Rare, crypt abscesses/ulcers
Fistulas and strictures → transmural inflammation	Fistulas and stricture are rare
Rectal involvement → spared	Rectal involvement- always involved
ASCA+	P-ANCA+
Bloody diarrhea- less common	Bloody diarrhea- more common
Extraintestinal- Seen in both (arthritis, uveitis, erythema nodosum, primary sclerosing cholangitis)	Same
Imaging findings String sign on barium study (narrowed terminal ileum)	Imaging findings Lead pipe colon on barium enema (loss of haustra)



HAMARTOMATOUS POLYP

1. JUVENILE POLYP

Age <5 yr

Mutation- SMAD2, SMAD4

c/f- bleeding per rectum

2. PEUTZ JEGHERS SYNDROME

Mutation - loss of function → STK 11/LKB 1 gene

Pigmentation on lips



Most common site - jejunum

Arborising polyp



3. COWDEN SYNDROME

PTEN gene mutation → chr 10

Syndrome include- PTEN

Polyps, breast cancer, Thyroid cancer, Trichilemmoma, Endometrial cancer

FAMILIAL ADENOMATOUS POLYPOSIS

No of polyps > 100

APC gene- chr5q21

Prophylactic colectomy does not eliminate the cancer

1. CLASSIC FAP

>100 polyps + congenital hypertrophy of RPE (retinal pigment epithelium)

2. TURCOT SYNDROME

FAP + malignant CNS tumor (eg, medulloblastoma, glioma)

3. GARDNER FAP

FAP with fibromatosis and osteomas + supernumerary teeth

LYNCH SYNDROME/HNPCC

Autosomal dominant mutation of Mismatched gene repair

Trick - CEO goes for lunch

FMGE 2025

C- proximal colon cancer
 E- endometrial cancer
 O-ovarian cancer

Hereditary hyperbilirubinemias

1.Gilbert syndrome

Mild deficiency of UGT

Asymptomatic or mild jaundice usually with stress, illness, or fasting

2.Dubin johnson syndrome

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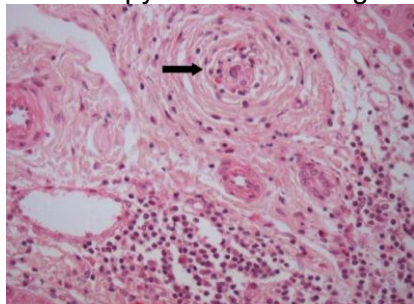
Conjugated bilirubin ↑ because the liver can't excrete it properly.

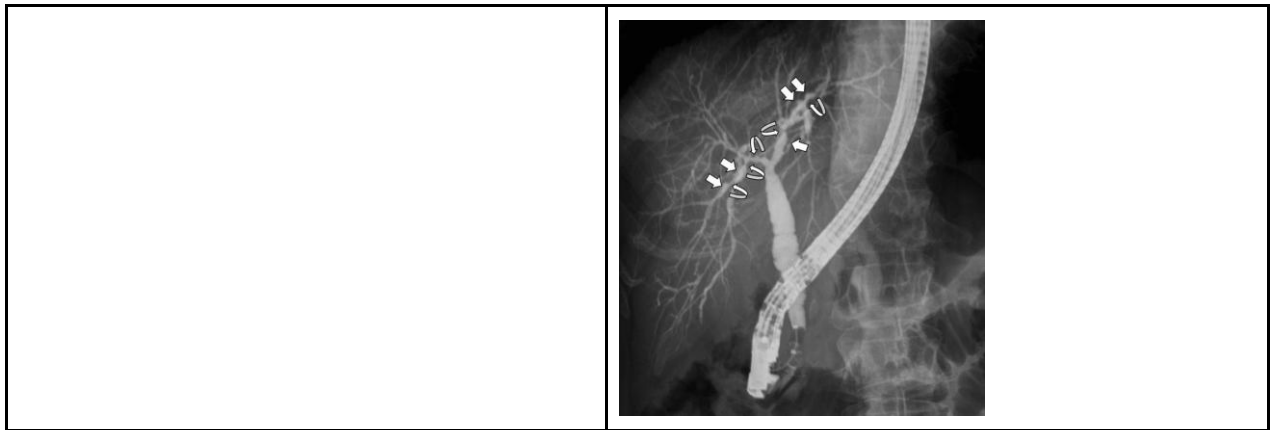
The liver looks dark/black due to buildup of certain waste products (epinephrine metabolites).

3.Rotor syndrome

Defect in organic anion transporter (OATP)

Due to impaired hepatic storage of conjugated bilirubin.

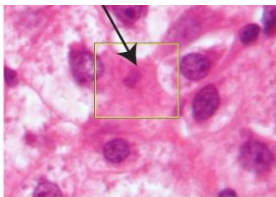
PRIMARY BILIARY CIRRHOSIS	PRIMARY SCLEROSING CHOLANGITIS
Autoimmune granulomatous destruction of intrahepatic bile ducts	Inflammation and fibrosis of intrahepatic and extrahepatic bile ducts
AMA+	p-ANCA +
Common in female	Common in male
Associated with sjogren syndrome	Associated with ulcerative colitis
Radiology - normal	Microscopy - onion skinning +  Radiology- pruning of small bile ducts, strictures and beading of large bile ducts



ACUTE HEPATITIS

FMGE 2023

Spotty necrosis and dropout necrosis
Councilman bodies are seen



Autoimmune Hepatitis

FMGE 2026

AIH 1	AIH 2	AIH 3
ANA - Antinuclear antibody	Anti LKM 1 - Hepatitis C	Anti SLA
ASMA - Anti smooth muscle antibody	Anti LKM2 - Drug induced Hepatitis	
SLA - Soluble liver antigen	Anti LKM3 - Hepatitis D	

HEPATITIS B

FMGE 2025,22

HbsAg → shows diffusely pink ground glass hepatocytes

CLASSICAL HCC

Age- >40yr

Associated with hepatitis b and c

Tumor marker → α FP

FIBROLAMELLAR VARIANT OF HCC

Age-20 yrs

Gene mutation → deletion 19

Tumor marker → α FP are normal, neurotensin level ↑

Micronodular cirrhosis

Granular surface of liver

Size of granules <3 mm

Causes

ALD(Alcoholic liver disease)

ICC(indian childhood cirrhosis)

PBC(Primary Biliary Cirrhosis)

Hemochromatosis

Macronodular cirrhosis.

Size of granules >3 mm

Causes:

ALD late stage

Viral hepatitis

Wilson disease

Alcoholic liver disease

Stages of ALD

1)Steatosis or Fatty liver.

2) SteatoHepatitis (**Mallory Hyaline or Denk Bodies**) are seen

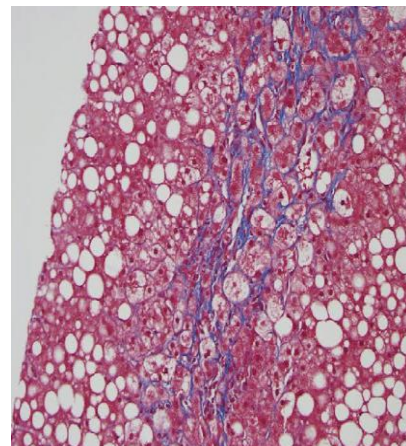
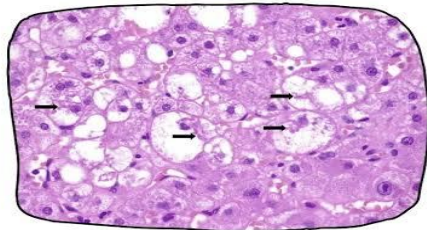
3) Cirrhosis

Chicken wire fibrosis

Mallory Denk bodies

(alcoholic hyaline/ mallory hyaline/ Eosinophilic bodies)

Seen inside hepatocytes, composed of intermediate filament (cytokeratin 8 & 18) & indicates endoplasmic reticulum stress.



Central stellate scar

FNH

FIBROLAMELLAR VARIANT OF HCC

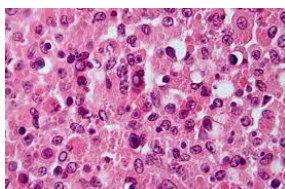
SEROUS CYSTADENOCARCINOMA

ONCOCYTOMA

CHROMOPHOBE RCC

NEETPG 2025

MALAKOPLAKIA



NEETPG 2022,24

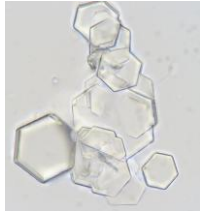
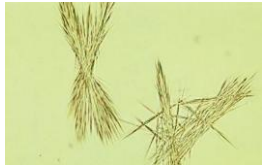
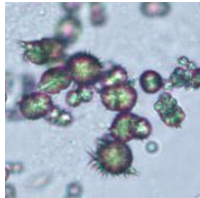
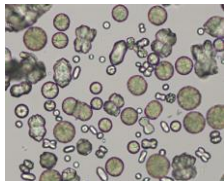
There is a phagolysosome defect (macrophages can't kill E. coli).
 This causes yellow-white plaques in mucosa that look like cancer.
 Calcium deposits form around E. coli, called Michaelis–Gutmann bodies.

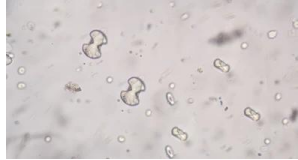
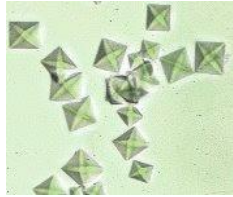
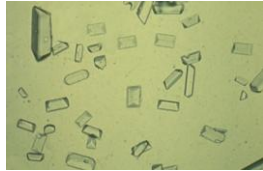
CAST

FMGE 2023

RBC cast - seen in glomerulonephritis 	WBC cast - chronic pyelonephritis 
Hyaline cast - made of tamm horsfall protein 	Muddy brown cast - seen in acute tubular necrosis 

CRYSTAL AND CASTS

Cysteine crystal - Hexagonal	
Tyrosine crystal - Thin needles	
Ammonium biurate crystal - Apple thorn app.	
Calcium carbonate crystal - Round wheel	

Calcium oxalate monohydrate crystal - Dumb bell	
Calcium oxalate dihydrate crystal - Envelope crystal	
Triple phosphate crystal/Struvite crystal - Coffin lid appearance Triple phosphate crystal/staghorn calculus associated with proteus Infection.	
Uric acid crystal - Rhombus, triangle, radiolucent	

NEPHRITIC SYNDROME

FMGE 2023, NEETPG 2023

1. Post streptococcal glomerulonephritis

Cause- strep pyogenes

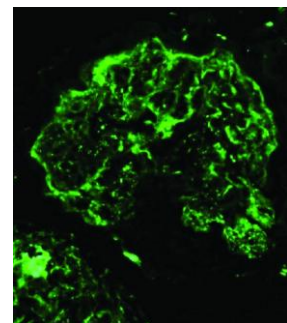
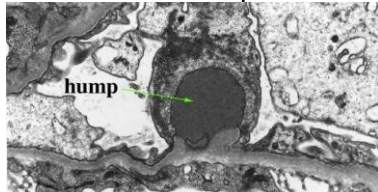
Sore throat → 2 weeks → cola color urine

Antibody - Anti DNase B

Hypocomplementemia

Immunofluorescence → starry sky appearance is seen due to deposition of Ag-Ab complex in some areas

Electron microscope → Subepithelial IC humps



2. IgA Nephropathy/Berger's disease

Sore throat → 2-3 days → recurrent hematuria

Cause: IgA immune complexes deposit in the kidney's mesangium.

Seen in children — causes blood in urine (red or brown urine) after infections like cold or diarrhea.

Reason: During infection, the body makes extra IgA → deposits in the kidney.

Seen on IF: IgA deposits in mesangium.

Can slowly lead to kidney failure.

3. Alport syndrome

Genetic problem in type IV collagen (mostly X-linked).

“Basket-weave” appearance due to irregular thickening and longitudinal splitting of GBM

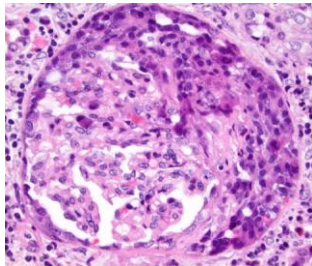
Triad- Anterior lenticonus, SNHL, renal failure

Trick → (can't see, can't pee, can't hear a bee)



Rapidly progressive glomerulonephritis

Composition → FLaP → fibrin, leukocytes, and parietal epithelium



Goodpasture syndrome

Trick- 1→2→3→4

Type I RPGN

Type II hypersensitivity reaction

Absent against $\alpha 3$ chain of type 4 collagen

c/f- Hemoptysis and hematuria

NEPHROTIC SYNDROME

NEETPG 2023

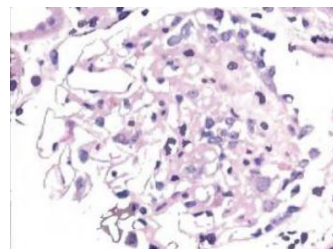
Focal segmental glomerulosclerosis

Findings:

Only some glomeruli and part of each glomerulus are scarred (focal segmental sclerosis).

Foot processes are lost on EM.

No immune deposits → negative IF.



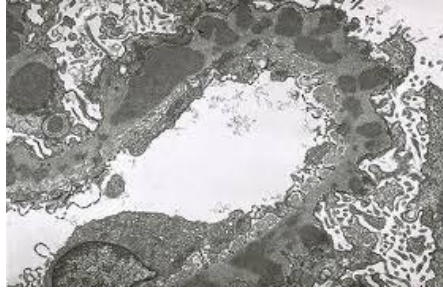
Membranous glomerulonephropathy

Cause: antibodies against phospholipase A2 receptor

(PLA2Rc), Hep B/C, solid tumors, SLE, or drugs (NSAIDs, penicillamine)

Findings →

Immune complex deposits under the epithelium → “spike and dome” look on EM.



Membranoproliferative Glomerulonephritis

Thick basement membrane with “tram-track” look.
Caused by immune complex deposits (granular IF).

Two types:

Type I: Deposits below endothelium → linked to Hep B & C.

Type II (dense deposit): Deposits inside membrane → due to C3 nephritic factor (↑ complement activity, ↓ C3).

Diabetes Nephropathy

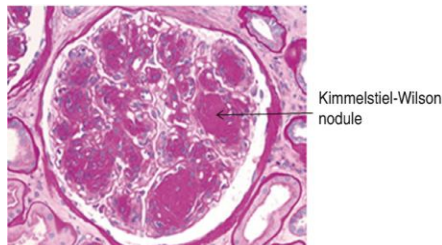
FMGE 2021

High blood sugar damages kidney vessels → hyaline arteriosclerosis.

Efferent arterioles are more affected → causes high pressure in glomerulus.

Leads to microalbuminuria → nephrotic syndrome.

Kimmelstiel–Wilson nodules seen in glomeruli.



THYROID CARCINOMA

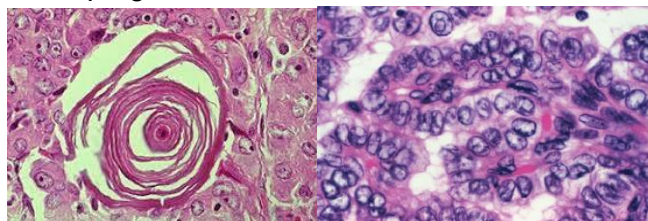
PAPILLARY CARCINOMA

FMGE 2023,21

Most common in childhood irradiation.

Orphan Annie eye nuclei and Psammoma bodies are seen.

Good prognosis



FOLLICULAR CARCINOMA

FMGE 2025, NEETPG 2021

Good prognosis

Invades thyroid capsule and vasculature

FNAC is not contraindicated → FNAC is avoided

Fine needle aspiration cytology may not be able to distinguish between follicular adenoma and carcinoma

MEDULLARY CARCINOMA

NEETPG 2025

Arise from Parafofollicular C cells and produce calcitonin

Associated with MEN 2A and 2B

Increase risk with RET oncogene

ANAPLASTIC CARCINOMA

Seen in an old aged patient.

Presents with rapidly enlarging neck mass Compressive symptoms (eg, dyspnea, dysphagia, hoarseness)

Poor prognosis and associated with TP53 mutation.

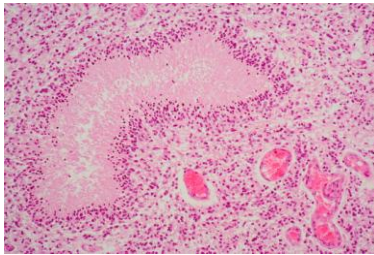
CNS TUMORS

GLIOBLASTOMA

Grade IV astrocytoma

Butterfly glioma seen

Pseudopalisading tumor cells seen



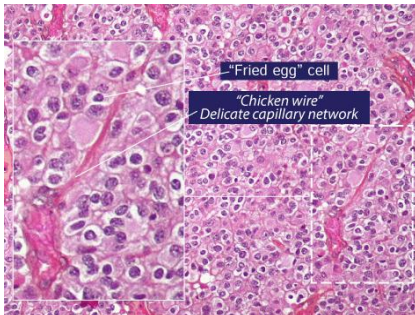
FMGE 2023,22



OLIGODENDROGLIOMA

Calcifications seen ‘

Fried egg colonies and chicken wire capillary pattern seen



MENINGIOMA

FMGE 2025.23

Common in female, it is benign

Extra-axial (external to brain parenchyma) and may have a dural attachment

Psammoma bodies

Calcification +



CRANIOPHARYNGIOMA

NEETPG 2025

Most common brain tumor in children (supratentorial).

It comes from Rathke's pouch (ectoderm).

Can look like a pituitary tumor → causes bitemporal hemianopia.

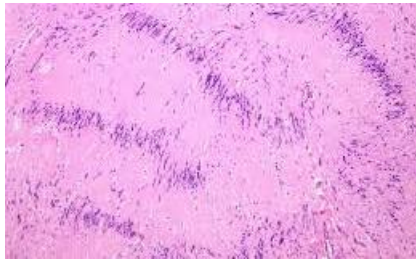
Calcified tumor with cholesterol crystals in "motor oil" fluid.

SCHWANNOMA

Seen at the cerebellopontine angle

Involve CN V, VII, and VIII

Bilateral vestibular schwannomas found in NF-2



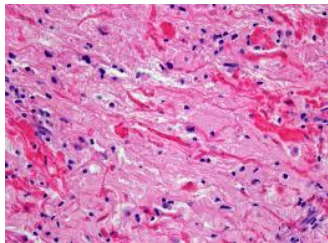
Verocay bodies

PILOCYTIC ASTROCYTOMA

Low grade astrocytoma

MC primary brain tumor in children

Rosenthal fibers (eosinophilic, corkscrew fibers)



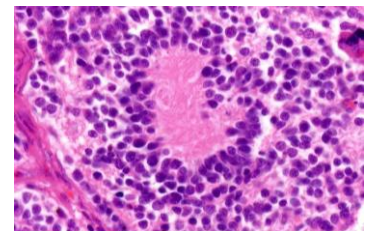
MEDULLOBLASTOMA

MC malignant brain tumor in childhood

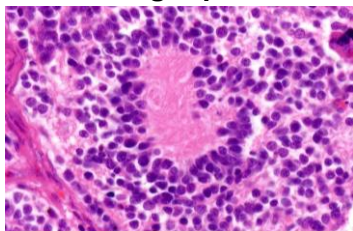
Involves cerebellum, Can compress 4th ventricle Involve the cerebellar vermis → truncal ataxia.

Drop metastases to the spinal cord.

Homer-Wright rosettes

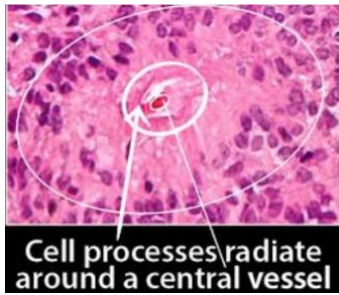


Homer wright pseudorosette- M/E/N



Medulloblastoma, Ewing sarcoma, Neuroblastoma

Perivascular pseudorosettes → blood vessel in the center → Ependymoma



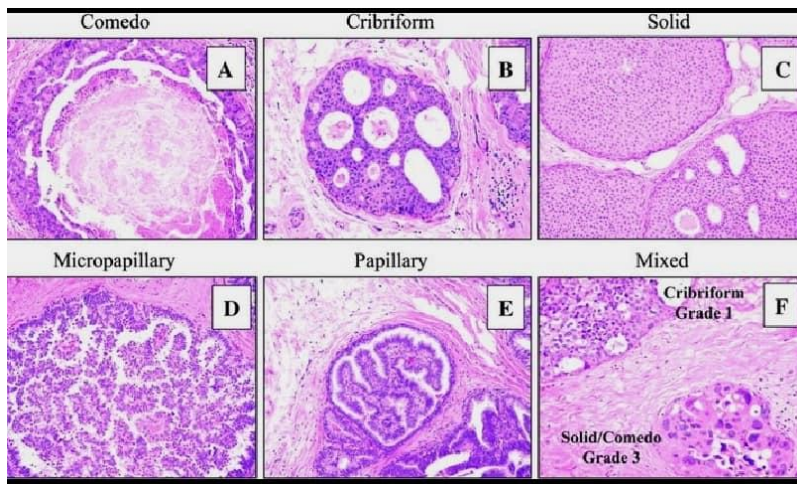
Flexner wintersteiner rosette– Retinoblastoma

FMGE 2024



BREAST

Lobular carcinoma insitu

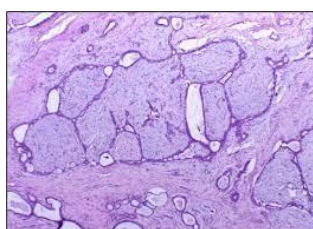
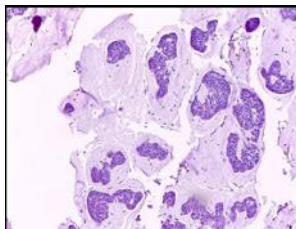


Mucinous carcinoma

Fibroadenoma

Phyllodes tumor

FMGE 2025,24



PAGET DISEASE OF BREAST

Skin finding → crusting,excoriation,pruritis

Biopsy shows → empty cells in epithelium

Hormone profile→ ER -ve,PR -ve , Her2/Neu +

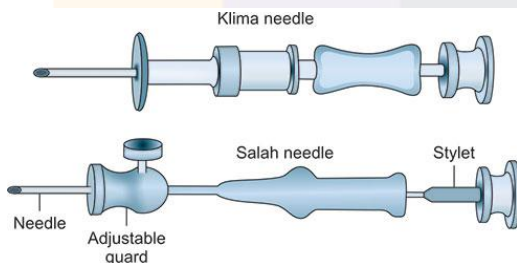
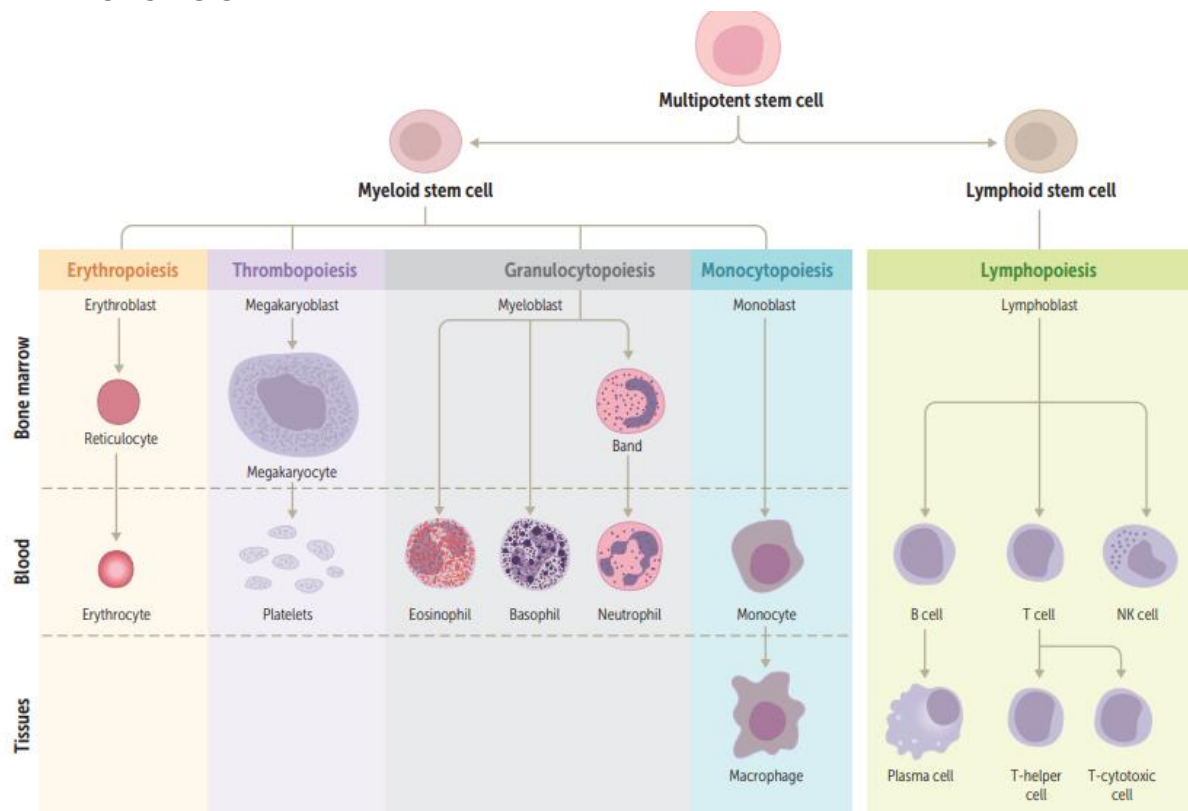


LUMINAL A	LUMINAL B	Her2 rich	Triple negative
ER → +ve	ER → +ve	ER → -ve	ER → -ve
PR→ +ve	PR→ +ve	PR→ -ve	PR→ -ve
Her2/neu→ -ve	Her2/neu→ -ve (sometime +ve)	Her2/neu→ +ve	Her2/neu→ -ve

Ki67 <14% → luminal A

Ki67 >14% → luminal B

HEMATOPOIESIS



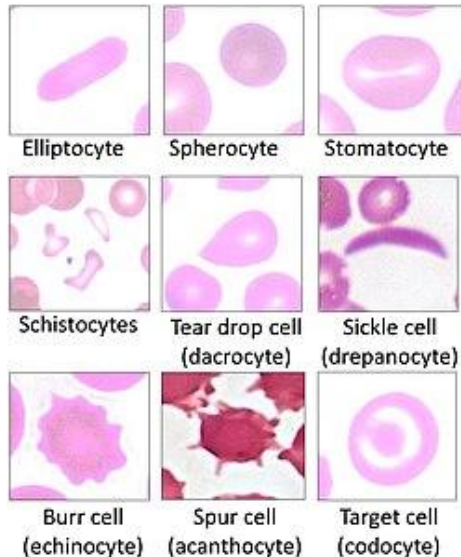
Bone marrow biopsy → Jamshidi needle



MICROCYTIC HYPOCHROMIC ANEMIA

Trick- SITA
 S- sideroblastic anemia
 I- IDA
 T- Thalessemia
 A- Anemia of chronic disease

POIKILOCYTOSIS

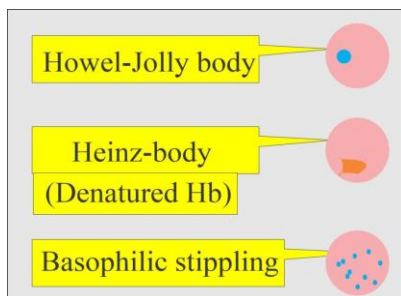


Acanthocytes ("spur cells") - Abetalipoproteinemia, vitamin E deficiency
 Dacrocytes ("teardrop cells") - Myelofibrosis
 Schistocytes ("helmet" cells) - MAHAs (eg, DIC, TTP/HUS, HELLP syndrome), mechanical hemolysis (eg, heart valve prosthesis)
 Degmacytes ("bite cells") - Degmacytes
 Spherocytes - Hereditary spherocytosis, autoimmune hemolytic anemia
 Target cells - HbC disease, Asplenia, Thalassemia

RBC inclusions

Peripheral smear

1. Howell-Jolly bodies - Functional hyposplenism (eg, sickle cell disease), asplenia
2. Basophilic stippling - Sideroblastic anemia, thalassemia
4. Heinz bodies - G6PD deficiency



FEATURES	IDA	SIDEROBLASTIC	AOCD
SERUM IRON	↓	↑	↓
SERUM FERRITIN	↓	↑	↑
TRANSFERRIN SATURATION	↓	↑	↓
TIBC	↑	↓	↓

IRON DEFICIENCY ANEMIA

FMGE 2024,23,NEETPG 2022,21

IRON METABOLISM

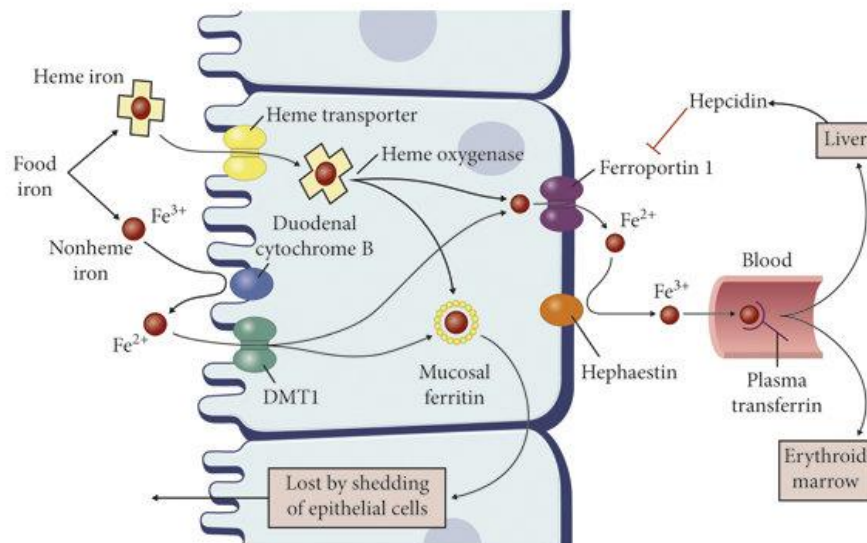
Site of absorption- Duodenum

Food contains iron in Fe^{3+}

Storage form of iron- Ferritin

Regulator of iron- Hepcidin

Iron transportation in form of - Transfer



Causes -

Duodenal disease

Inadequate iron diet

Parasitic infection

Iron studies

SERUM IRON	↓
SERUM FERRITIN	↓
TRANSFERRIN SATURATION	↓
TIBC	↑

Most specific test for IDA- bone marrow biopsy

ANEMIA OF CHRONIC DISEASE

Normocytic normochronic disease/microcytic hypochromic anemia

IL-6 release in chronic disease → ↑ hepcidin → ↓ iron absorption

Sideroblastic anemia

FMGE 2025

Iron overload condition

Causes:

Genetic: X-linked ALA synthase defect

Acquired: Myelodysplastic syndromes

Reversible: Alcohol, lead poisoning, ↓ vitamin B6, ↓ copper, drugs (isoniazid, linezolid)

Findings:

↑ Iron, normal/low TIBC, ↑ Ferritin

Ringed sideroblasts in bone marrow (iron around nucleus)

Basophilic stippling on blood smear

MACROCYTIC HYPOCHROMIC ANEMIA

1. Megaloblastic anemia

FMGE 2022

Problem: DNA makes slowly → nucleus matures late, cytoplasm normal.

Causes: Vitamin B12 deficiency, Folate deficiency, or drugs (hydroxyurea, phenytoin, methotrexate, sulfa), liver disease

Vitamin B12 (cobalamin) deficiency

NEETPG 2025,24

Site of absorption - ileum

B12 used for thymidine production → nuclear maturation

B12 helps in conversion of →

1. Methylmalonyl CoA to succinyl CoA

2. Homocysteine to methionine

→ B12 deficiency does not convert methylmalonyl CoA to succinyl CoA

↓ level of succinyl CoA → ↓ myelin production

→ B12 deficiency does not convert Homocysteine to methionine

↑ Homocysteine → ↑ atherosclerosis

Causes of B12 deficiency:

Pernicious anemia

Malabsorption (e.g. Crohn's disease)

Gastrectomy (no intrinsic factor)

vegan diet (low intake)

Fish tapeworm (*Diphyllobothrium latum*)

Findings -

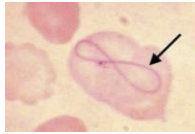
↑ homocysteine, ↑ methylmalonic acid.

B12 deficiency causes nerve problems — reversible dementia, loss of balance, and weakness (damage to spinal tracts), Subacute combined degeneration

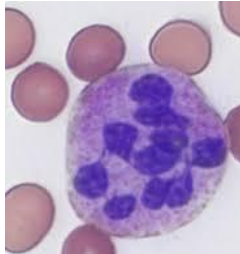
Folate fixes anemia but makes nerve damage worse.

Schilling test was used to find the cause (diet vs absorption)

RBC findings -
Howell jolly bodies +
Cabot ring +



WBC findings -
Hypersegmented neutrophils



FOLATE DEFICIENCY

Site of absorption - jejunum

Causes of folate deficiency:

Poor diet (alcoholism)

Drugs: methotrexate, trimethoprim, phenytoin

Increased need: pregnancy, hemolytic anemia

Findings-

↑ homocysteine, normal methylmalonic acid. No neurologic symptoms

FIGLU test → +

HEREDITARY SPHEROCYTOSIS

FMGE 2021, NEETPG 2023,24,25

Genetic (autosomal dominant) defect in RBC membrane proteins (ankyrin, spectrin, band 3, protein 4.2)

RBCs become small & round (no central pallor), Hb concentration ↑.

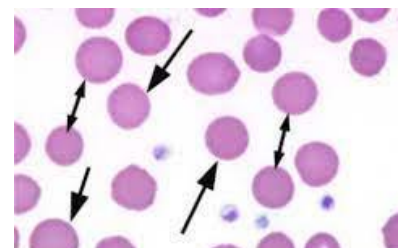
Water loss leads to reduction in size of the RBC.

Destroyed in spleen → anemia, splenomegaly, gallstones.

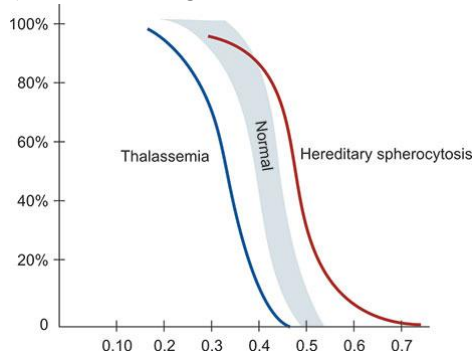
Crisis: Aplastic crisis with parvovirus B19.

Tests:

- ↑ Fragility in hypotonic solution (osmotic fragility test) → graph shift to right



- ↓ EMA binding test



Treatment: Splenectomy.

G6PD DEFICIENCY

X-linked recessive

↓ NADPH → ↓ reduced glutathione

↑ RBC susceptibility to oxidative stress (eg, sulfa drugs, antimalarials, fava beans)

→ Hemolysis.

Causes extravascular and intravascular hemolysis.

Confirmatory test - G6PD assay

PAROXYSMAL NOCTURNAL HEMOGLOBINURIA

NEETPG 2024

Only acquired intracorporeal defect

PIGA gene mutation → defective GPI anchor → loss of CD55 & CD59 (protective proteins).

Leads to complement mediated RBC destruction, especially at night.

C/F-

Hemolytic anemia

Thrombosis – most common cause of death

pancytopenia/aplastic anemia

Confirmatory test - FLOW CYTOMETRY

Treatment: eculizumab (targets terminal complement protein C5)

SICKLE CELL ANEMIA

FMGE 2024,23,NEETPG 2024

Missense Point mutation → at β subunit → glutamic acid change into valine

Low O₂, high altitude, or acidosis → HbS sickles → RBC destruction + vessel blockage.

Newborns: No symptoms (high HbF).

Carriers: Protected from malaria.

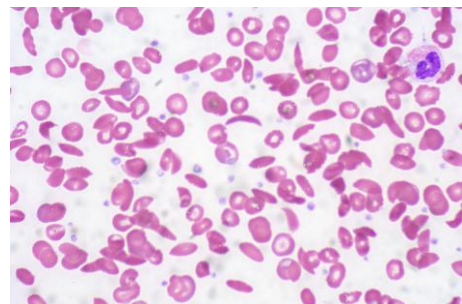
Findings: Sickle-shaped RBCs, crew-cut skull from ↑ bone marrow activity.

Types

HbAA - normal

HbAS - sickle cell trait

HbSS - sickle cell anemia



Complications:

- Aplastic crisis: Parvovirus B19 → ↓ RBCs
- Autosplenectomy: ↑ infection risk (esp. Salmonella)
- Splenic infarct / sequestration crisis
- Painful crises: dactylitis, priapism, acute chest syndrome, AVN, stroke
- Renal papillary necrosis → hematuria

Hb electrophoresis: ↓ HbA, ↑ HbF, ↑ HbS

Gold standard test → HPLC (high pressure liquid chromatography)

Treatment: Hydroxyurea ↑ HbF, hydration

THALASSEMIA

FMGE 2023

α thalassemia

Alpha gene deletion	state
(AA/AA)- all 4 genes present	Normal
(AA/A -) - 1 α gene deleted	Asymptomatic
(AA/- -) - 2 α gene deleted	α thalassemia trait
(A/- -) - 3 α gene deleted	Formation of β 4 tetramers Show golf ball inclusion HbH disease
(- -/- -) - all 4 α gene deleted	Hb barts disease Intrauterine death

β thalassemia

FMGE 2023, NEETPG 2022

Feature	Thalassemia major	Thalassemia intermediate	Thalassemia minor
Beta chain production	No β chain production	Little chain present	One β chain is normal
Iron profile	↑	Normal	Normal



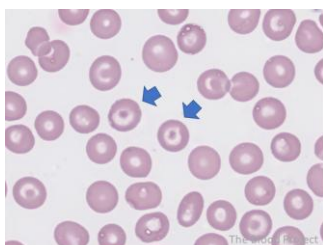
chipmunk facies



Hair on end appearance → seen

in both SCA and Thalassemia

Target cell or codocyte present



Confirmatory test - HPLC
NESTROF TEST - screening test



Mentzer index $\rightarrow$ MCV/RBC
 $>13 \rightarrow$ IDA
 $<13 \rightarrow$ Thalassemia

AUTOIMMUNE HEMOLYTIC ANEMIA

FMGE 2023

Types

1. Warm AIHA (37°C)

2. Cold AIHA (4°C)

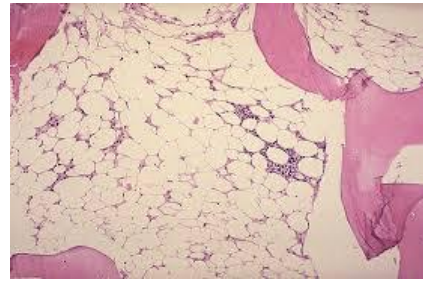
Warm AIHA	Cold AIHA
Antibody - IgG Temperature - 37°C IgG attaches to P antigen on the RBC $\rightarrow$ engulfed by spleen $\rightarrow$ causes extra vascular hemolysis Seen in CLL and SLE	IgM- cold agglutinin disease Temp $\rightarrow 4^{\circ}\text{C}$ Antibody $\rightarrow$ IgM IgM attaches to the I-antigen on the RBC IgM can activate the complement system and break off all the RBC to cause intravascular hemolysis. This does not take place because of the presence of CD55 and CD59 RBC coated with C3b changes into C3d Rbc taken up by liver and destroyed $\rightarrow$ extravascular hemolysis IgG cold AIHA Result in intravascular hemolysis

APLASTIC ANEMIA

FMGE 2021, NEETPG 2021

Causes -
Drugs, toxin
Viral infections

Pancytopenia →
Hb < 10gm
WBC < 4000
Plt < 1 lakh



Bone marrow aspirate → dry tap
Bone marrow biopsy → Investigation of choice

PURE RED CELL APLASIA

FMGE 2024, NEETPG 2021

Only RBC affected

Causes -

Parvovirus B19

Thymoma – associated with PRCA and Myasthenia gravis

CLL (CHRONIC LYMPHOCYTIC LEUKEMIA)

FMGE 2025,21, NEETPG 2023

Only leukemia not associated with radiation

Most common genetic defect → Gene deletion - 13q and 17q

Most common leukemia seen in elderly

C/F-

Lymphadenopathy, Hepatosplenomegaly

Diagnostic criteria

Blood → absolute lymphocyte count - >5000/mm³

Bone marrow → lymphocytic count >30%

CD5+ & CD23+ → FLOW CYTOMETRY → IOC

CD 200+

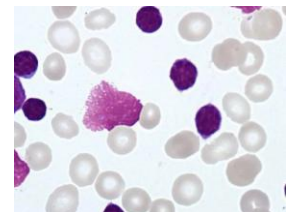
Smudge cell + due to loss of vimentin

CLL transformation

Convent school girl appearance

CLL → SLL → DLBCL (Richter syndrome)

Pseudofollicular pattern +



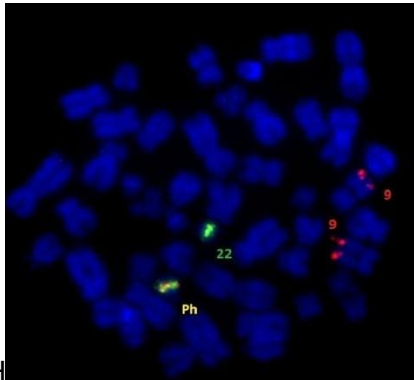
CML (CHRONIC MYELOID LEUKEMIA)

FMGE 2024,21, NEETPG 2021,22,24

t(9;22) - philadelphia chromosome

Chr 9 has ABL and Chr 22 has BCR
Fusion of ABL and BCR → ↑ tyrosine kinase activity

CML present with elevated WBC → Neutrophils, Metamyelocytes, Myelocytes, Eosinophils
SHIFT TO LEFT IS SEEN



Test - FISH

Trt- Imatinib

CML- phases → CAB

Chronic	<10% blast
Accelerated	10-19% Basophils >20% TLC ↑/↓
Blast crisis	≥20% blast

College girl appearance

AML (ACUTE MYELOID LEUKEMIA)

NEETPG 2023,25

Aetiology-

- BM failure • Defective DNA repair • Down syndrome
- MPO AND SUDAN BLACK B +

c/f-

- symptoms in less than 1 month
- stormy presentation fatigue, TLC inc/decrease, platelet decrease (purpura, epistaxis)

GINGIVAL HYPERPLASIA- M5 AML

Most common presentation DIC- t(15:17)

M3 AML IOC- BM aspiration → Auer rods +

Work up

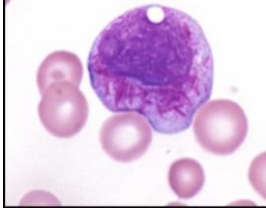
TLC increase/decrease

Platelet decrease

BM aspiration- by using Salah and Klima needle

AML- FAB classification

FMGE 2022, NEETPG 2024

AML M0	AML undifferentiated MPO negative
AML M1	AML without maturation
AML M2	AML with maturation -Most common t(8:21) RUNX1-RUNX1T1 fusion
AML M3	APML- acute promyelocytic leukemia t(15:17) → PML-RARA fusion PML-RARA- blocks promyelocyte to myelocyte transformation ↑ promyelocyte : criss-cross auer rods - faggot cells → contain procoagulant material  can causes- DIC Trt- ATRA- all trans retinoic acid + As2O3 - arsenic trioxide DOC → FISH
AML M4	Acute myelomonocytic leukemia Inv 16
AML M5	Acute monoblastic leukemia gingival hyperplasia
AML M6	Erythroleukemia
AML M7	Megakaryocytic leukemia- DOWN syndrome <3YRS

Bone marrow aspiration - dry tap

Dx- flow cytometry

ALL (ACUTE LYMPHOBLASTIC LEUKEMIA)

FMGE

2021

Most common leukemia - children

Most common in Down syndrome

Associated with CNS leukemia

B cell line- 76 percent

Common ALL- CD10 [t\(9:22\)](#)

T cell line - 24 percent

NOTCH mutation

Tissue infiltration

CNS

Mediastinal involvement

Testis +

Etiology

Philadelphia chromosome-
t(9:22), t(4:11), t(1:19), t(8:14)

C/F-

Sternal tenderness

Bone pain

Hepatosplenomegaly

Testicular mass

Workup

CBC-

- Hb decrease
- TLC normal/increase
- BMA- blast >20%

CD marker -

T - CD 1-8,28

B - CD 10,19,23,40

Genetics

Good prognosis	Bad prognosis
Hyperdiploidy → most common Trisomy 4,7,10 t(12:21) Female 2-10 yrs	Hypodiploidy t(9:22) MLL1 gene Male <2 - >10yrs

Flow cytometry → diagnosis of choice

HODGKIN LYMPHOMA

FMGE 2023

Cause- Epstein barr virus

Cell of origin - post germinal center B cell

HPE findings- owl eye appearance - Reed sternberg cells- CD15+

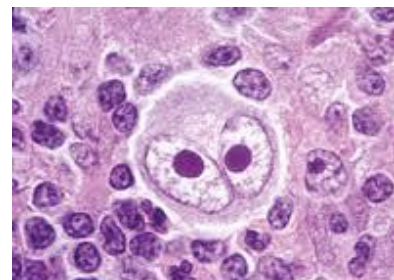
Marker - CD15,CD30 (most sensitive), PAX5 (most specific)

C/F-

- Progressive cervical lymphadenopathy
- Rubbery consistency LN
- Porta hepatis + LN+
- Fever +

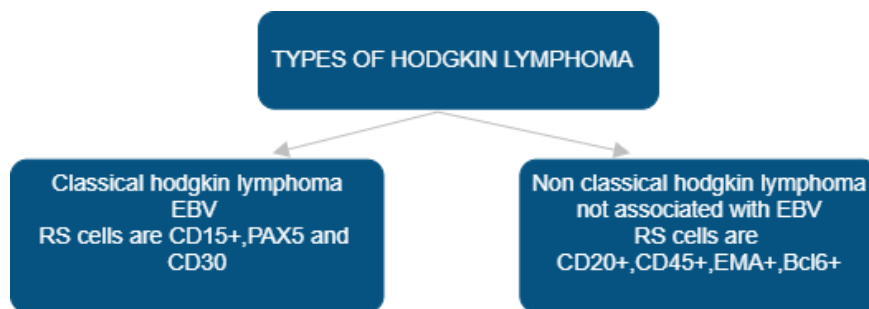
Work up

LDH increase



IOC- Excision LN biopsy

Flow Cytometry- CD15/30+



Types of classical hodgkin lymphoma

Classical hodgkin lymphoma	CD15,CD30,PAX5+ EBV +	
Nodular sclerosis	Lacunar cells	Mediastinal involvement Best prognosis
Mixed cellularity	Classical RS cells	Most common with HIV Most common with EBV Polymorphous cells seen(neutrophils,plasma cells,lymphocytes)
Lymphocyte rich	Mononuclear cells	
Lymphocyte depleted	Mummified RS cells	Associated with HIV Worst prognosis

Types of non classical hodgkin lymphoma

Non classical hodgkin lymphoma	Nodular lymphocyte predominant hodgkin lymphoma (NLPHL)
CD20,CD45,BCL6+ EBV -VE	Popcorn / L&H RS cells Best prognosis overall

NON HODGKIN LYMPHOMA

FMGE 2025

Chromosome 8 has c myc → burkitt lymphoma

Chromosome 11 & 18,t(11:18) → Marginal zone lymphoma

Chromosome 14 & 18,t(14:18) → Follicular lymphoma

Chromosome 11 & 14,t(11:14) → Mantle cell lymphoma

	CD 5	CD 23	CD 200
CLL/SLL	+	+	+
MCL (Mantle cell lymphoma)	+	-	-
MZL (Marginal zone lymphoma)	-	-	-

B-NHL

1. Burkitt lymphoma

More common in kid

t(8;14)—translocation of c-myc (8)

Associated with tumor lysis syndrome

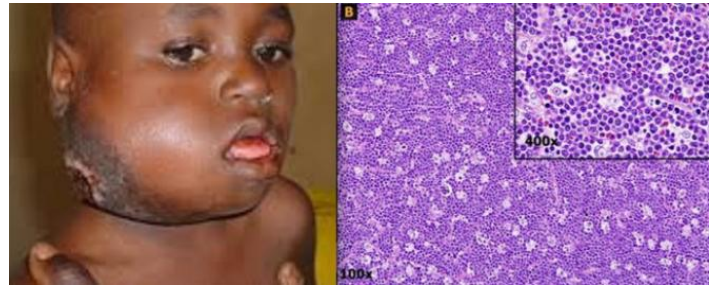
“Starry sky” appearance – lymphocytes with macrophages in between.

Linked to EBV infection.

Jaw tumor in African (endemic) type.

Ki67 → 100%

FMGE 2021, NEETPG 2023,25



2. Follicular Lymphoma

t(14:18)

Good prognosis

Marker → Bcl2+, CD19, CD20

3. Diffuse large B cell lymphoma (DLBCL)

Most common NHL in world

CLL/SLL → DLBCL → Richter syndrome

CD38+ and CD30+

4. Hairy cell leukemia

BRAFV600E mutation

Seen commonly in boys

Causes bone marrow fibrosis → dry tap on aspiration.

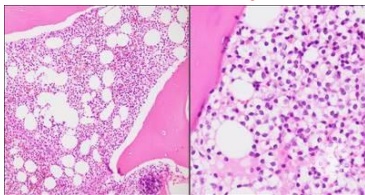
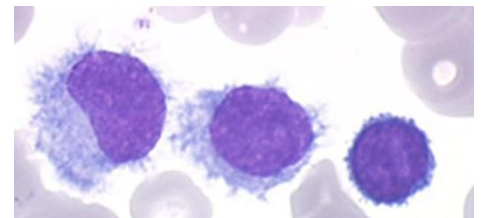
Massive splenomegaly and pancytopenia seen.

TRAP positive (Tartrate-Resistant Acid Phosphatase).

Marker → CD11c, CD25, CD103, ANNEXIN A1

Bone marrow biopsy → fried egg appearance

NEETPG 2024



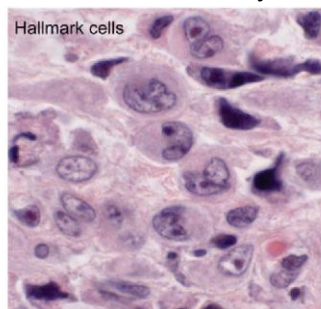
Diagnosis → by flow cytometry

T-NHL

1. Anaplastic large T cell lymphoma

Hallmark cell - kidney shaped cell → CD30+, t(2:5)

FMGE 2025



MULTIPLE MYELOMA

FMGE 2024,22,21,NEETPG 2021,24

Origin → post germinal centre B cell

Deletion 13q

t(11:14)

Overproduction of IgG (most common) > IgA

Abnormal plasma cell growth → Excess antibody production → These cells produce large amounts of **monoclonal immunoglobulin (M protein)** or **light chains (Bence Jones proteins)**.

Multiple myeloma in kidney → AL type light chain → cause CKD with secondary hyperparathyroidism → ↑ calcium

IL-6 is the main growth factor for myeloma cells → leads to bony lytic lesion

Criteria

Plasma cells >10% in bone marrow only

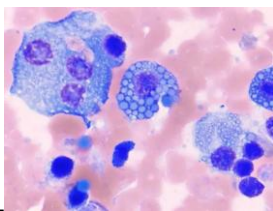
Ig>3gm/dl (IgG)

C- Calcium >11gm/dl

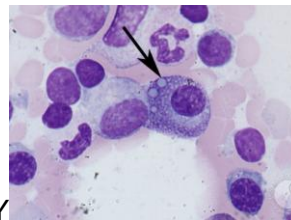
R- Renal insufficiency (s. Creatinine >2mg/dl)

A- Anemia (Hb<10gm)

B- Bony lytic lesion

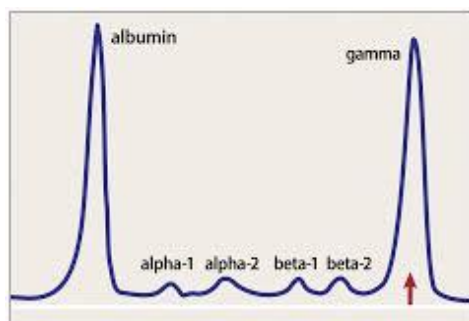


MOTT CELL



DUTCHER BODY

Serum protein electrophoresis



Abnormal result with myeloma cells producing the M-protein, creating an M-spike in the gamma region

Trt

Dexamethasone

Bortezomib

Lenalidomide

CLOT FORMATION

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Platelet adhesion → platelet secretion → platelet aggregation

Platelet adhesion

VWf release after endothelial injury bind to platelet with the help of Gp1b/IX

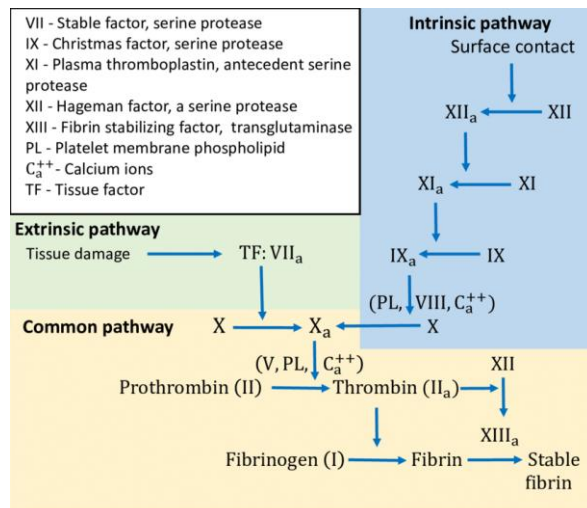
Platelet secretion

With the help of alpha and delta granules

Platelet aggregation

Gp2b/3a present on platelet help to adhere with another platelet

COAGULATION PATHWAY



PT- Extrinsic pathway- factor VII

aPTT- Intrinsic pathway- TENE- factor XII,XI,IX,VIII

Vitamin k dependent factors - II,VII,IX,X

HEMOPHILIA A, B, or C

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A: deficiency of factor VIII; X-linked recessive. → Trt → Cryoprecipitate (VWf and factor 8)

B: deficiency of factor IX; X-linked recessive. → Trt → fresh frozen plasma

C: deficiency of factor XI; autosomal recessive.

Von Willebrand factor

NEETPG 2024

VWD type 1	VWD type 2	VWD type 3
Most common Mild deficiency Quantitative defect	Qualitative defect Types 2a- ↓ in binding b/w VWF and Gp1b/9 2b- ↓ in binding b/w VWF and Gp1b/9	Most severe Quantitative defect
Trt- cryoprecipitate	cryoprecipitate	cryoprecipitate

PLATELET DISORDER

All platelet disorders have bleeding time (BT), mucous membrane bleeding, and microhemorrhages (eg, petechiae, epistaxis)

BERNARD- SOULIER SYNDROME

Defect in adhesion. decrease GpIb – decrease platelet-to-vWF adhesion.

Labs: decrease platelet aggregation

Big platelets.

GLANZMANN THROMBASTHENIA

Defect in aggregation.

decrease GpIIb/IIIa Decrease platelet-to-platelet aggregation and defective platelet plug formation

HEMOLYTIC UREMIC SYNDROME

Typical	Atypical
Associated with E coli Strain E coli → O157:H7	Factor H/CD46 deficiency due to complement dysregulation

Investigations:

↓ Hemoglobin

↓ Platelets

↑ Creatinine, ↑ BUN

Peripheral smear → schistocytes (fragmented RBCs)

Coombs test → negative

Disease	Platelet count	BT	PT	apTT
Bernard soulier syndrome	Normal	↑	Normal	Normal
Glanzmann thrombasthenia	Normal	↑	Normal	Normal
ITP	↓	↑	Normal	Normal
Hemophilia A	Normal	Normal	Normal	↑
Hemophilia B	Normal	Normal	Normal	↑
Von willebrand disease	Normal	↑	Normal	↑
HUS/TTP	↓	↑	Normal	Normal